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ENHANCED BILINGUAL EDITION

Original Arabic Terms Restored

THE ALGEBRA

of

MOHAMMED BEN MUSA

Kitāb al-Mukhtaṣar fī Ḥisāb al-Jabr wal-Muqābala

كتاب المختصر في حساب الجبر والمقابلة

EDITED AND TRANSLATED

BY

FREDERIC ROSEN

Enhanced Edition with Original Arabic Terminology

Six Canonical Cases in Bilingual Presentation

Geometrical Demonstrations with Arabic Labels

Complete Arabic–English Glossary

LONDON:
Printed for the Oriental Translation Fund,
And sold by
J. Murray, Albemarle Street;
Parbury, Allen, & Co., Leadenhall Street;
Thacker & Co., Calcutta; Treuttel & Wuertz, Paris;
And E. Fleischer, Leipzig.

1831

Preface to the Enhanced Edition

This enhanced edition of Frederic Rosen's classic 1831 translation restores the original Arabic terminology that Rosen rendered into English. Where Rosen translated al-Khwārizmī's technical terms into their English equivalents, this edition presents *both* the original Arabic *and* Rosen's English translation, enabling readers to appreciate the precise mathematical vocabulary of the Arabic tradition.

The Six Canonical Terms

Arabic	**Transliteration**	**Rosen's English**
مال	*māl*	"squares"
جذر	*jidhr*	"roots"
عدد	*'adad*	"numbers"
شيء	*shay'*	"thing" (the unknown)
الجبر	*al-jabr*	"restoration/completion"
المقابلة	*al-muqābala*	"reduction/opposition"

The Arabic title of the work is:

كتاب المختصر في حساب الجبر والمقابلة

Throughout this enhanced edition:

- Arabic terms are presented [alongside Rosen's English translation on first occurrence in each section.
- The six canonical cases of equations are given with bilingual headings.
- Geometrical demonstrations include the original Arabic labels.
- A complete Arabic–English glossary is appended.

The goal is to honour both al-Khwārizmī's original mathematical vocabulary and Rosen's pioneering English translation, presenting them together for the benefit of historians of mathematics, Arabists, and mathematicians alike.

The Algebra
of
Mohammed ben Musa

Edited and Translated
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Frederic Rosen

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Printed by J. L. Cox, Great Queen Street, London.

Preface

In the study of history, the attention of the observer is drawn by a peculiar charm towards those epochs, at which nations, after having secured their independence externally, strive to obtain an inward guarantee for their power, by acquiring eminence as great in science and in every art of peace as they have already attained in the field of war. Such an epoch was, in the history of the Arabs, that of the Caliphs Al Mansur, Harun al Rashid, and Al Mamun, illustrious contemporaries of Charlemagne; to the glory of which era, the volume now offered to the public, a new monument is endeavoured to be raised.

Abu Abdallah Mohammed ben Musa, of Khowarezm, who it appears, from his preface, wrote this Treatise at the command of the Caliph Al Mamun, was for a long time considered as the original inventor of Algebra.* That he was not the inventor of the Art, is now well established; but that he was the first Mohammedan who wrote upon it, is to be found asserted in several Oriental writers. Haji Khalfa, in his bibliographical work, cites the initial words of the treatise now before us,* and states, in two distinct passages, that its author, Mohammed ben Musa, was the first Mussulman who had ever written on the solution of problems by the rules of completion and reduction. Two marginal notes in the Oxford manuscript—from which the text of the present edition is taken—and an anonymous Arabic writer, whose Bibliotheca Philosophorum is frequently quoted by Casiri,* likewise maintain that this production of Mohammed ben Musa was the first work written on the subject* by a Mohammedan.

From the manner in which our author, in his preface, speaks of the task he had undertaken, we cannot infer that he claimed to be the inventor. He says that the Caliph Al Mamun encouraged him to write a popular work on Algebra: an expression which would seem to imply that other treatises were then already extant.

From a formula for finding the circumference of the circle, which occurs in the work itself (Text p. 51, Transl. p. 72), I have, in a note, drawn the conclusion, that part of the information comprised in this volume was derived from an Indian source; a conjecture which is supported by the direct assertion of the author of the *Bibliotheca Philosophorum* quoted by Casiri (i. 426, 428). That Mohammed ben Musa was conversant with Hindu science, is further evident from the fact* that he abridged, at Al Mamun's request—but before the accession of that prince to the caliphat—the Sindhind, or astronomical tables,

*"*Haec ars olim Mahomete, Mosis Arabis filio, initium sumsit: etenim hujus rei locuples testis Leo-nardus Pisanus.*" Such are the words with which Hieronymus Cardanus commences his *Ars Magna*, in which he frequently refers to the work here translated, in a manner to leave no doubt of its identity.

*I am indebted to the kindness of my friend Mr. Gustav Fluegel of Dresden, for a most interesting extract from this part of Haji Khalfa's work. Complete manuscript copies of the كشف الظنون are very scarce. The only two which I have hitherto had an opportunity of examining (the one bought in Egypt by Dr. Ehrenberg, and now deposited in the Royal Library at Berlin—the other among Rich's collection in the British Museum) are only abridgments of the original compilation, in which the quotation of the initial words of each work is generally omitted. The prospect of an edition and Latin translation of the complete original work, to be published by Mr. Fluegel, under the auspices of the Oriental Translation Committee, must under such circumstances be most gratifying to all friends of Asiatic Literature.

*ابن الابي written in the twelfth century. Bibliotheca Arabica Escurialensis, t. i. 426, 428.

*The first of these marginal notes stands at the top of the first page of the manuscript, and reads thus: هذا اول كتاب صنفه بيد مسلم في الجبر والمقابلة "This is the first book written on (the art of calculating by) completion and reduction by a Mohammedan: on this account the author has introduced into it rules of various kinds, in order to render useful the very rudiments of Algebra." The other scholium stands farther on: it is the same to which I have referred in my notes to the Arabic text, p. 177.

*Related by Ebn al Adami in the preface to his astronomical tables. Casiri, l. 427, 428. Colebrooke, Dissertation, &c. p. lxiv, lxxii.

translated by Mohammed ben Ibrahim al Fazari from the work of an Indian astronomer who visited the court of Almansur in the 156th year of the Hejira (A.D. 773).

The science as taught by Mohammed ben Musa, in the treatise now before us, does not extend beyond quadratic equations, including problems with an affected square. These he solves by the same rules which are followed by Diophantus,[*] and which are taught, though less comprehensively, by the Hindu mathematicians.[*] That he should have borrowed from Diophantus is not at all probable; for it does not appear that the Arabs had any knowledge of Diophantus' work before the middle of the fourth century after the Hejira, when Abu'l-Wafa Buzjani rendered it into Arabic.[*]

It is far more probable that the Arabs received their first knowledge of Algebra from the Hindus, who furnished them with the decimal notation of numerals, and with various important points of mathematical and astronomical information.

But under whatever obligation our author may be to the Hindus, as to the subject matter of his performance, he seems to have been independent of them in the manner of digesting and treating it: at least the method which he follows in expounding his rules, as well as in showing their application, differs considerably from that of the Hindu mathematical writers. Bhaskara and Brahmagupta give dogmatical precepts, unsupported by argument, which, even by the metrical form in which they are expressed, seem to address themselves rather to the memory than to the reasoning faculty of the learner: Mohammed gives his rules in simple prose, and establishes their accuracy by geometrical illustrations. The Hindus give comparatively few examples, and are fond of investing the statement of their problems in rhetorical pomp: the Arab, on the contrary, is remarkably rich in examples, but he introduces them with the same perspicuous simplicity of style which distinguishes his rules. In solving their problems, the Hindus are satisfied with pointing at the result, and at the principal intermediate steps which lead to it: the Arab shows the working of each example at full length, keeping his view constantly fixed upon the two sides of the equation, as on the two scales of a balance, and showing how any alteration in one side is counterpoised by a corresponding change in the other.

Besides the few facts which have already been mentioned in the course of this preface, little or nothing is known of our Author's life. He lived and wrote under the caliphat of Al Mamun, and must therefore be distinguished from Abu Jafar Mohammed ben Musa,[*] likewise a mathematician and astronomer, who flourished under the Caliph Al Motaded

[*]See Diophantus, Introd. § ii. and Book iv. problems 32 and 33.

[*]Lilavati, p. 29, Vijaganita, p. 347, of Mr. Colebrooke's translation.

[*]Casiri Bibl. Arab. Escur. i. 433. Colebrooke's Dissertation, &c. p. lxxii.

[*]The father of the latter, Musa ben Shaker, whose native country I do not find recorded, had been a robber or bandit in the earlier part of his life, but had afterwards found means to attach himself to the court of the Caliph Al-Mamun; who, after Musa's death, took care of the education of his three sons, Mohammed, Ahmed, and Al Hassan. (Abulfaragii Histor. Dyn. p. 280. Casiri, l. 386, 418). Each of the sons subsequently distinguished himself in mathematics and astronomy. We learn from Abulfaraj (p. 281) and from Ibn Khallikan (art. ثابت بن قرة) that Thabet ben Korrah, the well-known translator of the Almagest, was indebted to Mohammed for his introduction to Al Motaded, and the men of science at the court of that caliph. Ibn Khallikan's words are: خرج من حران وسكن كفرطابا فاقام بها الى ان قدم عليه محمد بن موسى راجعا من بلاد الروم الى بغداد. فصحبه واجتهد في معرفته وعرف فضله وذكاءه فحمله الى بغداد فنزله داره وقربه الى المتوكل وكان من جملة اهل الحجاب "(Thabet ben Korrah) left Harran, and established himself at Kafratutha, where he remained till Mohammed ben Musa arrived there, on his return from the Greek dominions to Bagdad. The latter became acquainted with Thabet and on seeing his skill and sagacity, invited Thabet to accompany him to Bagdad, where Mohammed made him lodge at his own house, introduced him to the Caliph, and procured him an appointment in the body of astronomers." Ibn Khallikan here speaks of Mohammed ben Musa as of a well-known individual: he has however devoted no special article to an account of his life. It is possible that the tour into the provinces of the Eastern Roman Empire here mentioned, was undertaken in search of some ancient Greek works on mathematics or astronomy.

(who reigned A.H. 279–289, A.D. 892–902).

The manuscript from whence the text of the present edition is taken—and which is the only copy the existence of which I have as yet been able to trace—is preserved in the Bodleian collection at Oxford. It is, together with three other treatises on Arithmetic and Algebra, contained in the volume marked cmxviii. Hunt. 214, fol., and bears the date of the transcription A.H. 743 (A.D. 1342). It is written in a plain and legible hand, but unfortunately destitute of most of the diacritical points: a deficiency which has often been very sensibly felt; for though the nature of the subject matter can but seldom leave a doubt as to the general import of a sentence, yet the true reading of some passages, and the precise interpretation of others, remain involved in obscurity. Besides, there occur several omissions of words, and even of entire sentences; and also instances of words or short passages written twice over, or words foreign to the sense introduced into the text. In printing the Arabic part, I have included in brackets many of those words which I found in the manuscript, the genuineness of which I suspected, and also such as I inserted from my own conjecture, to supply an apparent hiatus.

The margin of the manuscript is partially filled with scholia in a very small and almost illegible character, a few specimens of which will be found in the notes appended to my translation. Some of them are marked as being extracted from a commentary (شرح) by Al Mozaihafi,* probably the same author, whose full name is Jamaleddin Abu Abdallah Mohammed ben Omar al Jaza'if al Mozaihafi, and whose "Introduction to Arithmetic," (مبادي علم الحساب) is contained in the same volume with Mohammed's work in the Bodleian library.

Numerals are in the text of the work always expressed by words: figures are only used in some of the diagrams, and in a few marginal notes.

The work had been only briefly mentioned in Uris' catalogue of the Bodleian manuscripts. Mr. H. T. Colebrooke first introduced it to more general notice, by inserting a full account of it, with an English translation of the directions for the solution of equations, simple and compound, into the notes of the "Dissertation" prefixed to his invaluable work, "Algebra, with Arithmetic and Mensuration, from the Sanscrit of Brahmegupta and Bhascara." (London, 1817, 4to. pages lxxv–lxxix.)

The account of the work given by Mr. Colebrooke excited the attention of a highly distinguished friend of mathematical science, who encouraged me to undertake an edition and translation of the whole: and who has taken the kindest interest in the execution of my task. He has with great patience and care revised and corrected my translation, and has furnished the commentary, subjoined to the text, in the form of common algebraic notation. But my obligations to him are not confined to this only; for his luminous advice has enabled me to overcome many difficulties, which, to my own limited proficiency in mathematics, would have been almost insurmountable.

In some notes on the Arabic text which are appended to my translation, I have endeavoured, not so much to elucidate, as to point out for further enquiry, a few circumstances connected with the history of Algebra. The comparisons drawn between the Algebra of the Arabs and that of the early Italian writers might perhaps have been more numerous and more detailed; but my enquiry was here restricted by the want of some important works. Montucla, Cossali, Hutton, and the Basil edition of Cardanus' *Ars magna*, were the only sources which I had the opportunity of consulting.

*Wherever I have met with this name, it is written without the diacritical points المزيحفي, and my pronunciation rests on mere conjecture.

The Author's Preface

In the Name of God, gracious and merciful!

This work was written by Mohammed ben Musa, of Khowarezm. He commences it thus:

Praised be God for his bounty towards those who deserve it by their virtuous acts: in performing which, as by him prescribed to his adoring creatures, we express our thanks, and render ourselves worthy of the continuance (of his mercy), and preserve ourselves from change: acknowledging his might, bending before his power, and revering his greatness! He sent Mohammed (on whom may the blessing of God repose!) with the mission of a prophet, long after any messenger from above had appeared, when justice had fallen into neglect, and when the true way of life was sought for in vain. Through him he cured of blindness, and saved through him from perdition, and increased through him what before was small, and collected through him what before was scattered. Praised be God our Lord! and may his glory increase, and may all his names be hallowed—besides whom there is no God; and may his benediction rest on Mohammed the Prophet and on his descendants!

The learned in times which have passed away, and among nations which have ceased to exist, were constantly employed in writing books on the several departments of science and on the various branches of knowledge, bearing in mind those that were to come after them, and hoping for a reward proportionate to their ability, and trusting that their endeavours would meet with acknowledgment, attention, and remembrance—content as they were even with a small degree of praise; small, if compared with the pains which they had undergone, and the difficulties which they had encountered in revealing the secrets and obscurities of science.

Some applied themselves to obtain information which was not known before them, and left it to posterity; others commented upon the difficulties in the works left by their predecessors, and defined the best method (of study), or rendered the access (to science) easier or placed it more within reach; others again discovered mistakes in preceding works, and arranged that which was confused, or adjusted what was irregular, and corrected the faults of their fellow-labourers, without arrogance towards them, or taking pride in what they did themselves.

That fondness for science, by which God has distinguished the Imam al Mamun, the Commander of the Faithful (besides the caliphat which He has vouchsafed unto him by lawful succession, in the robe of which He has invested him, and with the honours of which He has adorned him), that affability and condescension which he shows to the learned, that promptitude with which he protects and supports them in the elucidation of obscurities and in the removal of difficulties,—has encouraged me to compose a short work on Calculating by (the rules of) Completion and Reduction, confining it to what is easiest and most useful in arithmetic, such as men constantly require in cases of inheritance, legacies, partition, law-suits, and trade, and in all their dealings with one another, or where the measuring of lands, the digging of canals, geometrical computation, and other objects of various sorts and kinds are concerned—relying on the goodness of my intention therein, and hoping that the learned will reward it, by obtaining (for me) through their prayers the excellence of the Divine mercy: in requital of which, may the choicest blessings and the abundant bounty of God be theirs! My confidence rests with God, in this as in every thing, and in Him I put my trust. He is the Lord of the Sublime Throne. May His blessing descend upon all the prophets and heavenly messengers!

Mohammed ben Musa's

Compendium

on Calculating by

Completion and Reduction.

Kitāb al-Mukhtaṣar fī Ḥisāb al-Jabr wal-Muqābala

كتاب المختصر في حساب الجبر والمقابلة

Introduction

When I considered what people generally want in calculating, I found that it always is a number. I also observed that every number is composed of units, and that any number may be divided into units. Moreover, I found that every number, which may be expressed from one to ten, surpasses the preceding by one unit: afterwards the ten is doubled or tripled, just as before the units were: thus arise twenty, thirty, &c., until a hundred; then the hundred is doubled and tripled in the same manner as the units and the tens, up to a thousand; then the thousand can be thus repeated at any complex number; and so forth to the utmost limit of numeration.

I observed that the numbers which are required in calculating by Completion and Reduction [الجبر والمقابلة are of three kinds, namely, **roots [جذر**, **squares [مال**, and **simple numbers relative to neither root nor square [عدد**.

A *root* [جذر is any quantity which is to be multiplied by itself, consisting of units, or numbers ascending, or fractions descending.*

A *square* [مال is the whole amount of the root multiplied by itself.

A *simple number* [عدد is any number which may be pronounced without reference to root or square.

A number belonging to one of these three classes may be equal to a number of another class; you may say, for instance, "*squares are equal to roots*," or "*squares are equal to numbers*," or "*roots are equal to numbers*."

Case I.—Squares are equal to Roots. [المال يساوي الجذور

Of the case in which squares are equal to roots, this is an example: "A square is equal to five roots of the same;" the root of the square is five, and the square is twenty-five, which is equal to five times its root.

So you say, "one third of the square is equal to four roots;" then the whole square is equal to twelve roots; that is a hundred and forty-four; and its root is twelve.

Or you say, "five squares are equal to ten roots;" then one square is equal to two roots; the root of the square is two, and its square is four.

*By the word root, is meant the simple power of the unknown quantity.

In this manner, whether the squares be many or few, (i.e. multiplied or divided by any number), they are reduced to a single square; and the same is done with the roots, which are their equivalents; that is to say, they are reduced in the same proportion as the squares.

Case II.—Squares are equal to Numbers. **[**المال يساوي الأعداد

As to the case in which squares are equal to numbers; for instance, you say, "a square is equal to nine;" then this is a square, and its root is three. Or "five squares are equal to eighty;" then one square is equal to one-fifth of eighty, which is sixteen. Or "the half of the square is equal to eighteen;" then the square is thirty-six, and its root is six.

Thus, all squares, multiples, and sub-multiples of them, are reduced to a single square. If there be only part of a square, you add thereto, until there is a whole square; you do the same with the equivalent in numbers.

Case III.—Roots are equal to Numbers. **[**الجذور تساوي الأعداد

As to the case in which roots are equal to numbers; for instance, "one root equals three in number;" then the root is three, and its square nine. Or "four roots are equal to twenty;" then one root is equal to five, and the square to be formed of it is twenty-five. Or "half the root is equal to ten;" then the whole root is equal to twenty, and the square which is formed of it is four hundred.

I found that these three kinds; namely, **roots [**جذور, **squares [**أموال, and **numbers [**أعداد, may be combined together, and thus three compound species arise;[*] that is, "*squares and roots equal to numbers*;" "*squares and numbers equal to roots*;" "*roots and numbers equal to squares*."

Case IV.—Roots and Squares are equal to Numbers. **[**الجذور والمال يساويان الأعداد

Roots and Squares are equal to Numbers;[*] for instance, "one square, and ten roots of the same, amount to thirty-nine dirhems;" that is to say, what must be the square which, when increased by ten of its own roots, amounts to thirty-nine? The solution is this: you halve the number[*] of the roots, which in the present instance yields five. This you multiply by itself; the product is twenty-five. Add this to thirty-nine; the sum is sixty-four. Now take the root of this, which is eight, and subtract from it half the number of the roots, which is five; the remainder is three. This is the root of the square which you sought for; the square itself is nine.

The solution is the same when two squares or three, or more or less be specified;[*] you reduce them to one single square, and in the same proportion you reduce also the roots

[*] The three cases considered are,

1^{st}. $cx^2 + bx = a$

2^{d}. $cx^2 + a = bx$

3^{d}. $cx^2 = bx + a$

[*] 1^{st} case: $cx^2 + bx = a$.
Example: $x^2 + 10x = 39$
$x = \sqrt{25 + 39} - 5 = \sqrt{64} - 5 = 8 - 5 = 3$
i.e. the coefficient.

[*] By "the number of the roots" is meant the coefficient.

[*] $cx^2 + bx = a$ is to be reduced to the form $x^2 + \frac{b}{c}x = \frac{a}{c}$.

and simple numbers which are connected therewith.

For instance, “two squares and ten roots are equal to forty-eight dirhems;”* that is to say, what must be the amount of two squares which, when summed up and added to ten times the root of one of them, make up a sum of forty-eight dirhems? You must at first reduce the two squares to one; and you know that one square of the two is the moiety of both. Then reduce every thing mentioned in the statement to its half, and it will be the same as if the question had been, a square and five roots of the same are equal to twenty-four dirhems; or, what must be the amount of a square which, when added to five times its root, is equal to twenty-four dirhems? Now halve the number of the roots; the moiety is two and a half. Multiply that by itself; the product is six and a quarter. Add this to twenty-four; the sum is thirty dirhems and a quarter. Take the root of this; it is five and a half. Subtract from this the moiety of the number of the roots, that is two and a half; the remainder is three. This is the root of the square, and the square itself is nine.

The proceeding will be the same if the instance be, “half of a square and five roots are equal to twenty-eight dirhems;”* that is to say, what must be the amount of a square, the moiety of which, when added to the equivalent of five of its roots, is equal to twenty-eight dirhems? Your first business must be to complete your square, so that it amounts to one whole square. This you effect by doubling it. Therefore double it, and double also that which is added to it, as well as what is equal to it. Then you have a square and ten roots, equal to fifty-six dirhems. Now halve the roots; the moiety is five. Multiply this by itself; the product is twenty-five. Add this to fifty-six; the sum is eighty-one. Extract the root of this; it is nine. Subtract from this the moiety of the number of roots, which is five; the remainder is four. This is the root of the square which you sought for; the square is sixteen, and half the square eight.

Proceed in this manner, whenever you meet with squares and roots that are equal to simple numbers: for it will always answer.

Case V.—Squares and Numbers are equal to Roots. [المال والأعداد يساويان الجذور

Squares and Numbers are equal to Roots;* for instance, “a square and twenty-one in numbers are equal to ten roots of the same square.” That is to say, what must be the amount of a square, which, when twenty-one dirhems are added to it, becomes equal to the equivalent of ten roots of that square?

Solution: Halve the number of the roots; the moiety is five. Multiply this by itself; the product is twenty-five. Subtract from this the twenty-one which are connected with the square; the remainder is four. Extract its root; it is two. Subtract this from the moiety of the roots, which is five; the remainder is three. This is the root of the square which you required, and the square is nine. Or you may add the root to the moiety of the roots; the sum is seven; this is the root of the square which you sought for, and the square itself is forty-nine.

When you meet with an instance which refers you to this case, try its solution by addition, and if that do not serve, then subtraction certainly will. For in this case both addition and subtraction may be employed, which will not answer in any other of the

* $2x^2 + 10x = 48$, reduced to $x^2 + 5x = 24$,
$x = \sqrt{6\frac{1}{4} + 24} - 2\frac{1}{2} = \sqrt{30\frac{1}{4}} - 2\frac{1}{2} = 5\frac{1}{2} - 2\frac{1}{2} = 3.$

* $\frac{1}{2}x^2 + 10x = 56$; $x^2 + 10x = 56$; $x = \sqrt{25 + 56} - 5 = \sqrt{81} - 5 = 9 - 5 = 4.$

* 2^d case: $cx^2 + a = bx$.
Example: $x^2 + 21 = 10x$
$x = \frac{10}{2} \pm \sqrt{(\frac{10}{2})^2 - 21} = 5 \pm \sqrt{25 - 21} = 5 \pm \sqrt{4} = 5 \pm 2 = 7$ or 3.

three cases in which the number of the roots must be halved.[*] And know, that, when in a question belonging to this case you have halved the number of the roots and multiplied the moiety by itself, if the product be less than the number of dirhems connected with the square, then the instance is impossible;[*] but if the product be equal to the dirhems by themselves, then the root of the square is equal to the moiety of the roots alone, without either addition or subtraction.

In every instance where you have two squares, or more or less, reduce them to one entire square,[*] as I have explained under the first case.

Case VI.—Roots and Numbers are equal to Squares. [الجذور والأعداد تساوي المال]

Roots and Numbers are equal to Squares;[*] for instance, "three roots and four of simple numbers are equal to a square." Solution: Halve the roots; the moiety is one and a half. Multiply this by itself; the product is two and a quarter. Add this to the four; the sum is six and a quarter. Extract its root; it is two and a half. Add this to the moiety of the roots, which was one and a half; the sum is four. This is the root of the square, and the square is sixteen.

Whenever you meet with a multiple or sub-multiple of a square, reduce it to one entire square.

These are the six cases which I mentioned in the introduction to this book. They have now been explained. I have shown that three among them do not require that the roots be halved, and I have taught how they must be resolved. As for the other three, in which halving the roots is necessary, I think it expedient, more accurately, to explain them by separate chapters, in which a figure will be given for each case, to point out the reasons for halving.

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[*] If in an equation, of the form $x^2 + a = bx$, $(\frac{b}{2})^2 < a$, the case supposed in the equation cannot happen. If $(\frac{b}{2})^2 = a$, then $x = \frac{b}{2}$.

[*] The case of $(\frac{b}{2})^2 < a$ gives no real solution.

[*] $cx^2 + a = bx$ is to be reduced to $x^2 + \frac{a}{c} = \frac{b}{c}x$.

[*] 3^d case: $cx^2 = bx + a$.

Example: $x^2 = 3x + 4$

$x = \sqrt{(\frac{3}{2})^2 + 4} + \frac{3}{2} = \sqrt{2\frac{1}{4} + 4} + 1\frac{1}{2} = \sqrt{6\frac{1}{4}} + 1\frac{1}{2} = 2\frac{1}{2} + 1\frac{1}{2} = 4.$

Geometrical Demonstrations

Demonstration of the Case: *"a Square* [مال *and ten Roots* [جذور *are equal to thirty-nine Dirhems* [درهم."[*]

The figure to explain this a quadrate, the sides of which are unknown. It represents the square, the root of which, or the root of which, you wish to know. This is the figure AB, each side of which may be considered as one of its roots; and if you multiply one of these sides by any number, then the amount of that number may be looked upon as the number of the roots which are added to the square. Each side of the quadrate represents the root of the square; and, as in the instance the roots were connected with the square, we may take one-fourth of ten, that is to say, two and a half, and combine it with each of the four sides of the figure. Thus with the original quadrate AB, four new parallelograms are combined, each having a side of the quadrate as its length, and the number of two and a half as its breadth; they are the parallelograms C, G, T, and K. We have now a quadrate of equal, though unknown sides; but in each of the four corners of which a square piece of two and a half multiplied by two and a half is wanting. In order to compensate for this want and to complete the quadrate, we must add (to that which we have already) four times the square of two and a half, that is, twenty-five. We know (by the statement) that the first figure, namely, the quadrate representing the square, together with the four parallelograms around it, which represent the ten roots, is equal to thirty-nine of numbers. If to this we add twenty-five, which is the equivalent of the four quadrates at the corners of the figure AB, by which the great figure DH is completed, then we know that this together makes sixty-four. One side of this great quadrate is its root, that is, eight. If we subtract twice a fourth of ten, that is five, from eight, as from the two extremities of the side of the great quadrate DH, then the remainder of such a side will be three, and that is the root of the square, or the side of the original figure AB.

It must be observed, that we have halved the number of the roots, and added the product of the moiety multiplied by itself to the number thirty-nine, in order to complete the great figure in its four corners; because the fourth of any number multiplied by itself, and then by four, is equal to the product of the moiety of that number multiplied by itself.[*] Accordingly, we multiplied only the moiety of the roots by itself, instead of multiplying its fourth by itself, and then by four.

This is the figure:

	G	
C	AB	T
	K	

The same may also be explained by another figure. We proceed from the quadrate AB, which represents the square. It is our next business to add to it the ten roots of the same. We halve for this purpose the ten, so that it becomes five, and construct two quadrangles on two sides of the quadrate AB, namely, G and D, the length of each of them being five, as the moiety of the ten roots, whilst the breadth of each is equal to a side of the quadrate AB. Then a quadrate remains opposite the corner of the quadrate AB. This is equal to five multiplied by five: this five being half of the number of the roots which we have added to each of the two sides of the first quadrate. Thus we know that the first quadrate, which is the square, and the two quadrangles on its sides, which are the ten

[*] Geometrical illustration of the case, $x^2 + 10x = 39$.

[*] $4 \times (\frac{b}{4})^2 = (\frac{b}{2})^2$.

roots, make together thirty-nine. In order to complete the great quadrate, there wants only a square of five multiplied by five, or twenty-five. This we add to thirty-nine, in order to complete the great square SH. The sum is sixty-four. We extract its root, eight, which is one of the sides of the great quadrangle. By subtracting from this the same quantity which we have before added, namely five, we obtain three as the remainder. This is the side of the quadrangle AB, which represents the square; it is the root of this square, and the square itself is nine.

This is the figure:

D	
AB	G

Demonstration of the Case: "a Square and twenty-one Dirhems are equal to ten Roots."

We represent the square by a quadrate AD, the length of whose side we do not know. To this we join a parallelogram, the breadth of which is equal to one of the sides of the quadrate AD, such as the side HN. This parallelogram is HB. The length of the two figures together is equal to the line HC. We know that its length is ten of numbers; for every quadrate has equal sides and angles, and one of its sides multiplied by a unit is the root of the quadrate, or multiplied by two it is twice the root of the same. As it is stated, therefore, that a square and twenty-one of numbers are equal to ten roots, we may conclude that the length of the line HC is equal to ten of numbers, since the line CD represents the root of the square. We now divide the line CH into two equal parts at the point G: the line GC is then equal to HG. It is also evident that the line GT is equal to the line CD. At present we add to the line GT, in the same direction, a piece equal to the difference between CG and GT, in order to complete the square. Then the line TK becomes equal to KM, and we have a new quadrate of equal sides and angles, namely, the quadrate MT. We know that the line TK is five; this is consequently the length also of the other sides: the quadrate itself is twenty-five, this being the product of the multiplication of half the number of the roots by themselves, for five times five is twenty-five.

We have perceived that the quadrangle HB represents the twenty-one of numbers which were added to the quadrate. We have then cut off a piece from the quadrangle HB by the line KT (which is one of the sides of the quadrate MT), so that only the part TA remains. At present we take from the line KM the piece KL, which is equal to GK; it then appears that the line TG is equal to ML; moreover, the line KL, which has been cut off from KM, is equal to KG; consequently, the quadrangle MR is equal to TA. Thus it is evident that the quadrangle HT, augmented by the quadrangle MR, is equal to the quadrangle HB, which represents the twenty-one.

The whole quadrate MT was found to be equal to twenty-five. If we now subtract from this quadrate, MT, the quadrangles HT and MR, which are equal to twenty-one, there remains a small quadrate KR, which represents the difference between twenty-five and twenty-one. This is four; and its root, represented by the line RG, which is equal to GA, is two. If you subtract this number two from the line CG, which is the moiety of the roots, then the remainder is the line AC; that is to say, three, which is the root of the original square. But if you add the number two to the line CG, which is the moiety of the number of the roots, then the sum is seven, represented by the line CR, which is the root to a larger square. However, if you add twenty-one to this square, then the sum will likewise be equal to ten roots of the same square.

Demonstration of the Case: "three Roots and four of Simple Numbers are equal to a Square."

Let the square be represented by a quadrangle, the sides of which are unknown to us, though they are equal among themselves, as also the angles. This is the quadrate AD, which comprises the three roots and the four of numbers mentioned in this instance.

In every quadrate one of its sides, multiplied by a unit, is its root. We now cut off the quadrangle HD from the quadrate AD, and take one of its sides HC for three, which is the number of the roots. The same is equal to RD. It follows, then, that the quadrangle HB represents the four of numbers which are added to the roots. Now we halve the side CH, which is equal to three roots, at the point G; from this division we construct the square HT, which is the product of half the roots (or one and a half) multiplied by themselves, that is to say, two and a quarter. We add then to the line GT a piece equal to the line AH, namely, the piece TL; accordingly the line GL becomes equal to AG, and the line KN equal to TL. Thus a new quadrangle, with equal sides and angles, arises, namely, the quadrangle GM and we find that the line AG is equal to ML, and the same line AG is equal to GL. By these means the line CG remains equal to NR, and the line MN equal to TL, and from the quadrangle HB a piece equal to the quadrangle KL is cut off.

But we know that the quadrangle AR represents the four of numbers which are added to the three roots. The quadrangle AN and the quadrangle KL are together equal to the quadrangle AR, which represents the four of numbers.

We have seen, also, that the quadrangle GM comprises the product of the moiety of the roots, or of one and a half, multiplied by itself; that is to say two and a quarter, together with the four of numbers, which are represented by the quadrangles AN and KL.

There remains now from the side of the great original quadrate AD, which represents the whole square, only the moiety of the roots, that is to say, one and a half, namely, the line GC. If we add this to the line AG, which is the root of the quadrate GM, being equal to two and a half; then this, together with CG, or the moiety of the three roots, namely, one and a half, makes four, which is the line AC, or the root to a square, which is represented by the quadrate AD.

Here follows the figure. This it was which we were desirous to explain.

We have observed that every question which requires equation or reduction for its solution, will refer you to one of the six cases which I have proposed in this book. I have now also explained their arguments. Bear them, therefore, in mind.

On Multiplication

I SHALL NOW TEACH YOU how to multiply the unknown numbers, that is to say, the roots, one by the other, if they stand alone, or if numbers are added to them, or if numbers are subtracted from them, or if they are subtracted from numbers; also how to add them one to the other, or how to subtract one from the other.

Whenever one number is to be multiplied by another, the one must be repeated as many times as the other contains units.[*]

If there are greater numbers combined with units to be added to or subtracted from them, then four multiplications are necessary;[*] namely, the greater numbers by the greater numbers, the greater numbers by the units, the units by the greater numbers, and the units by the units.

If the units, combined with the greater numbers, are positive, then the last multiplication is positive; if they are both negative, then the fourth multiplication is likewise positive. But if one of them is positive, and one negative, then the fourth multiplication is negative.[*]

For instance, “ten and one to be multiplied by ten and two.”[*] Ten times ten is a hundred; once ten is ten positive; twice ten is twenty positive, and once two is two positive; this altogether makes a hundred and thirty-two.

But if the instance is “ten less one, to be multiplied by ten less one,”[*] then ten times ten is a hundred; the negative one by ten is ten negative; the other negative one by ten is likewise ten negative, so that it becomes eighty: but the negative one by the negative one is one positive, and this makes the result eighty-one.

Or if the instance be “ten and two, to be multiplied by ten less one,”[*] then ten times ten is a hundred, and the negative one by ten is ten negative; the positive two by ten is twenty positive; this together is a hundred and ten; the positive two by the negative one gives two negative. This makes the product a hundred and eight.

I have explained this, that it might serve as an introduction to the multiplication of unknown sums, when numbers are added to them, or when numbers are subtracted from them, or when they are subtracted from numbers.

For instance: “Ten less **thing [**شيء □□□□□□□□, the signification of **root [**جذر □□□□□□□□, to be multiplied by ten.”[*] You begin by taking ten times ten, which is a hundred; less thing by ten is ten roots negative; the product is therefore a hundred less ten things.

If the instance be: “ten and thing to be multiplied by ten,”[*] then you take ten times ten, which is a hundred, and thing by ten is ten things positive; so that the product is a hundred plus ten things.

If the instance be: “ten and thing to be multiplied by itself,”[*] then ten times ten is a hundred, and ten times thing is ten things; and again, ten times thing is ten things; and

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[*]If x is to be multiplied by y, x is to be repeated as many times as there are units in y.

[*]If $x \pm a$ is to be multiplied by $y \pm b$, x is to be multiplied by y, x is to be multiplied by b, a is to be multiplied by y, and a is to be multiplied by b.

[*]In multiplying $y \pm x = a \pm b$:
$(+a)(+b) = +ab$; $(-a)(-b) = +ab$; $(+a)(-b) = -ab$; $(-a)(+b) = -ab$.

[*]$(10 + 1) \times (10 + 2) = 10 \times 10 + 1 \times 10 + 2 \times 10 + 1 \times 2 = 100 + 10 + 20 + 2 = 132.$

[*]$(10 - 1) \times (10 - 1) = 10 \times 10 - 10 - 10 + 1 = 100 - 20 + 1 = 81.$

[*]$(10 + 2) \times (10 - 1) = 10 \times 10 - 10 + 20 - 2 = 100 - 10 + 20 - 2 = 108.$

[*]$(10 - x) \times 10 = 10 \times 10 - 10x = 100 - 10x.$

[*]$(10 + x) \times 10 = 10 \times 10 + 10x = 100 + 10x.$

[*]$(10 + x)(10 + x) = 10 \times 10 + 10x + 10x + x^2 = 100 + 20x + x^2.$

thing multiplied by thing is a square positive, so that the whole product is a hundred dirhems and twenty things and one positive square.

If the instance be: "ten minus thing to be multiplied by ten minus thing," then ten times ten is a hundred; and minus thing by ten is minus ten things; and again, minus thing by ten is minus ten things. But minus thing multiplied by minus thing is a positive square. The product is therefore a hundred and a square, minus twenty things.

In like manner if the following question be proposed to you: "one dirhem minus one-sixth to be multiplied by one dirhem minus one-sixth;" that is to say, five-sixths by themselves, the product is five and twenty parts of a dirhem, which is divided into six and thirty parts, or two-thirds and one-sixth of a sixth. Computation: You multiply one dirhem by one dirhem, the product is one dirhem; then one dirhem by minus one-sixth, that is one-sixth negative; then, again, one dirhem by minus one-sixth is one-sixth negative: so far, then, the result is two-thirds of a dirhem: but there is still minus one-sixth to be multiplied by minus one-sixth, which is one-sixth of a sixth positive; the product is, therefore, two-thirds and one sixth of a sixth.

If the instance be, "ten minus thing to be multiplied by ten and thing," then you say,[*] ten times ten is a hundred; and minus thing by ten is ten things negative; and thing by ten is ten things positive; and minus thing by thing is a square positive; therefore, the product is a hundred dirhems, minus a square.

If the instance be, "ten minus thing to be multiplied by thing," then you say, ten multiplied by thing is ten things; and minus thing by thing is a square negative; therefore, the product is ten things minus a square.

If the instance be, "ten and thing to be multiplied by thing less ten," then you say, thing multiplied by ten is ten things positive; and thing by thing is a square positive; and minus ten by ten is a hundred dirhems negative; and minus ten by thing is ten things negative. You say, therefore, a square minus a hundred dirhems; for, having made the reduction, that is to say, having removed the ten things positive by the ten things negative, there remains a square minus a hundred dirhems.

If the instance be, "ten dirhems and half a thing to be multiplied by half a dirhem, minus five things," then you say, half a dirhem by ten is five dirhems positive; and half a dirhem by half a thing is a quarter of thing positive; and minus five things by ten dirhems is fifty roots negative. This altogether makes five dirhems minus forty-nine things and three quarters of thing. After this you multiply five roots negative by half a root positive: it is two squares and a half negative. Therefore, the product is five dirhems, minus two squares and a half, minus forty-nine roots and three quarters of a root.

If the instance be, "ten and thing to be multiplied by thing less ten," then this is the same as if it were said thing and ten by thing less ten. You say, therefore, thing multiplied by thing is a square positive; and ten by thing is ten things positive; and minus ten by thing is ten things negative. You now remove the positive by the negative, then there only remains a square. Minus ten multiplied by ten is a hundred, to be subtracted from the square. This, therefore, altogether, is a square less a hundred dirhems.

Whenever a positive and a negative factor concur in a multiplication, such as thing positive and minus thing, the last multiplication gives always the negative product. Keep this in memory.

[*] $(10-x)(10+x) = 10 \times 10 - 10x + 10x - x^2 = 100 - x^2$.

On Addition and Subtraction

KNOW THAT the root of two hundred minus ten, added to twenty minus the root of two hundred, is just ten.*

The root of two hundred, minus ten, subtracted from twenty minus the root of two hundred, is thirty minus twice the root of two hundred; twice the root of two hundred is equal to the root of eight hundred.*

A hundred and a square minus twenty roots, added to fifty and ten roots minus two squares, is a hundred and fifty, minus a square and minus ten roots.*

A hundred and a square, minus twenty roots, diminished by fifty and ten roots minus two squares, is fifty dirhems and three squares minus thirty roots.*

I shall hereafter explain to you the reason of this by a figure, which will be annexed to this chapter.

If you require to double the root of any known or unknown square, (the meaning of its duplication being that you multiply it by two) then it will suffice to multiply two by two, and then by the square;* the root of the product is equal to twice the root of the original square.

If you require to take it thrice, you multiply three by three, and then by the square; the root of the product is thrice the root of the original square.

Compute in this manner every multiplication of the roots, whether the multiplication be more or less than two.

If you require to find the moiety of the root of the square, you need only multiply a half by a half, which is a quarter; and then this by the square: the root of the product will be half the root of the first square. Follow the same rule when you seek for a third, or a quarter of a root, or any larger or smaller quota of it, whatever may be the denominator or the numerator.

Examples of this: If you require to double the root of nine,* you multiply two by two, and then by nine: this gives thirty-six; take the root of this, it is six, and this is double the root of nine.

In the same manner, if you require to triple the root of nine,* you multiply three by three, and then by nine; the product is eighty-one; take its root, it is nine, which becomes equal to thrice the root of nine.

If you require to have the moiety of the root of nine,* you multiply a half by a half, which gives a quarter, and then this by nine; the result is two and a quarter: take its root; it is one and a half, which is the moiety of the root of nine.

You proceed in this manner with every root, whether positive or negative, and whether known or unknown.

* $\sqrt{200} - 10 + (20 - \sqrt{200}) = 10.$
* $20 - \sqrt{200} - (\sqrt{200} - 10) = 30 - 2\sqrt{200} = 30 - \sqrt{800}.$
* $50 + 10x - 2x^2 + (100 + x^2 - 20x) = 150 - 10x - x^2.$
* $100 + x^2 - 20x - [50 - 2x^2 + 10x] = 50 + 3x^2 - 30x.$
* $(2)^2 \times x^2 = 4x^2$; therefore $\sqrt{4x^2} = 2x.$
* $3\sqrt{9} = \sqrt{9 \times 9} = \sqrt{81} = 9.$
* $3\sqrt{9} = \sqrt{9 \times 9} = \sqrt{81} = 9.$
* $\frac{1}{2}\sqrt{9} = \sqrt{\frac{1}{4} \times 9} = \sqrt{2\frac{1}{4}} = 1\frac{1}{2}.$

On Division

If you will divide the root of nine by the root of four,[*] you begin with dividing nine by four, which gives two and a quarter: the root of this is the number which you require—it is one and a half.

If you will divide the root of four by the root of nine,[*] you divide four by nine; it is four-ninths of the unit; the root of this is two divided by three; namely, two-thirds of the unit.

If you wish to divide twice the root of nine by the root of four, or of any other square,[*] you double the root of nine in the manner above shown to you in the chapter on Multiplication, and you divide the product by four, or by any number whatever. You perform this in the way above pointed out.

In like manner, if you wish to divide three roots of nine, or more, or one-half or any multiple or sub-multiple of the root of nine, the rule is always the same: follow it, the result will be right.

If you wish to multiply the root of nine by the root of four,[*] multiply nine by four; this gives thirty-six; take its root, it is six; this is the root of nine, multiplied by the root of four.

Thus, if you wish to multiply the root of five by the root of ten,[*] multiply five by ten: the root of the product is what you have required.

If you wish to multiply the root of one-third by the root of a half,[*] you multiply one-third by a half: it is one-sixth: the root of one-sixth is equal to the root of one-third, multiplied by the root of a half.

If you require to multiply twice the root of nine by thrice the root of four,[*] then take twice the root of nine, according to the rule above given, so that you may know the root of what square it is. You do the same with respect to the three roots of four in order to know what must be the square of such a root. You then multiply these two squares, the one by the other, and the root of the product is equal to twice the root of nine, multiplied by thrice the root of four.

You proceed in this manner with all positive or negative roots.

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[*] $\frac{\sqrt{9}}{\sqrt{4}} = \sqrt{\frac{9}{4}} = \frac{3}{2} = 1\frac{1}{2}$.

[*] $\frac{\sqrt{4}}{\sqrt{9}} = \sqrt{\frac{4}{9}} = \frac{2}{3}$.

[*] $\frac{2\sqrt{9}}{\sqrt{4}} = \frac{\sqrt{4}\times\sqrt{9}}{\sqrt{4}} = \sqrt{9} = 3$.

[*] $\sqrt{4} \times \sqrt{9} = \sqrt{4 \times 9} = \sqrt{36} = 6$.

[*] $\sqrt{10} \times \sqrt{5} = \sqrt{5 \times 10} = \sqrt{50}$.

[*] $\sqrt{\frac{1}{3}} \times \sqrt{\frac{1}{2}} = \sqrt{\frac{1}{3} \times \frac{1}{2}} = \sqrt{\frac{1}{6}}$.

[*] $3\sqrt{4} \times 2\sqrt{9} = \sqrt{9 \times 4} \times \sqrt{4 \times 9} = \sqrt{36 \times 36} = 36$.

Demonstrations

THE ARGUMENT for the root of two hundred, minus ten, added to twenty, minus the root of two hundred, may be elucidated by a figure.

Let the line AB represent the root of two hundred; let the part from A to the point C be the ten, then the remainder of the root of two hundred will correspond to the remainder of the line AB, namely to the line CB. Draw now from the point B a line to the point D, to represent twenty; let it, therefore, be twice as long as the line AC, which represents ten; and mark a part of it from the point B to the point H, to be equal to the line AB, which represents the root of two hundred; then the remainder of the twenty will be equal to the part of the line, from the point H to the point D.

As our object was to add the remainder of the root of two hundred, after the subtraction of ten, that is to say, the line CB, to the line HD, or to twenty, minus the root of two hundred, we cut off from the line BH a piece equal to CB, namely, the line SH. We know already that the line AB, or the root of two hundred, is equal to the line BH, and that the line AC, which represents the ten, is equal to the line SB, as also that the remainder of the line AB, namely, the line CB is equal to the remainder of the line BH, namely, to SH. Let us add, therefore, this piece SH, to the line HD. We have already seen that from the line BD, or twenty, a piece equal to AC, which is ten, was cut off, namely, the piece BS. There remains after this the line SD, which, consequently, is equal to ten.

This it was that we intended to elucidate.

The argument for the root of two hundred, minus ten, to be subtracted from twenty, minus the root of two hundred, is as follows.

Let the line AB represent the root of two hundred, and let the part thereof, from A to the point C, signify the ten mentioned in the instance. We draw now from the point B, a line towards the point D, to signify twenty. Then we trace from B to the point H, the same length as the length of the line which represents the root of two hundred; that is of the line AB. We have seen that the line CB is the remainder from the twenty, after the root of two hundred has been subtracted.

It is our purpose, therefore, to subtract the line CB from the line HD; and we now draw from the point B, a line towards the point S, equal in length to the line AC, which represents the ten. Then the whole line SD is equal to SB, plus BD, and we perceive that all this added together amounts to thirty.

We now cut off from the line HD, a piece equal to CB, namely, the line HG; thus we find that the line GD is the remainder from the line SD, which signifies thirty. We see also that the line BH is the root of two hundred and that the line SB and BC is likewise the root of two hundred. Now the line HG is equal to CB; therefore the piece subtracted from the line SD, which represents thirty, is equal to twice the root of two hundred, or once the root of eight hundred.

This it is that we wished to elucidate.

As for the hundred and square minus twenty roots added to fifty, and ten roots minus two squares, this does not admit of any figure, because there are three different species, viz. squares, and roots, and numbers, and nothing corresponding to them by which they might be represented. We had, indeed, contrived to construct a figure also for this case, but it was not sufficiently clear.

The elucidation by words is very easy. You know that you have a hundred and a square, minus twenty roots. When you add to this fifty and ten roots, it becomes a hundred and fifty and a square, minus ten roots. The reason for these ten negative roots is, that from the twenty negative roots ten positive roots were subtracted by reduction. This being done, there remains a hundred and fifty and a square, minus ten roots. With the hundred a square is connected. If you subtract from this hundred and square the two squares negative connected with fifty, then one square disappears by reason of the other, and the remainder is a hundred and fifty, minus a square, and minus ten roots.

This it was that we wished to explain.

Of the Six Problems

BEFORE THE CHAPTERS on computation and the several species thereof, I shall now introduce six problems, as instances of the six cases treated of in the beginning of this work. I have shown that three among these cases, in order to be solved, do not require that the roots be halved, and I have also mentioned that the calculating by completion and reduction must always necessarily lead you to one of these cases. I now subjoin these problems, which will serve to bring the subject nearer to the understanding, to render its comprehension easier, and to make the arguments more perspicuous.

First Problem.

I have divided ten into two portions; I have multiplied the one of the two portions by the other; after this I have multiplied the one of the two by itself, and the product of the multiplication by itself is four times as much as that of one of the portions by the other.*

Computation: Suppose one of the portions to be thing, and the other ten minus thing: you multiply thing by ten minus thing; it is ten things minus a square. Then multiply it by four, because the instance states "four times as much." The result will be four times the product of one of the parts multiplied by the other. This is forty things minus four squares. After this you multiply thing by thing, that is to say, one of the portions by itself. This is a square, which is equal to forty things minus four squares. Reduce it now by the four squares, and add them to the one square. Then the equation is: forty things are equal to five squares; and one square will be equal to eight roots, that is, sixty-four; the root of this is eight, and this is one of the two portions, namely, that which is to be multiplied by itself. The remainder from the ten is two, and that is the other portion. Thus the question leads you to one of the six cases, namely, that of "squares equal to roots." Remark this.

Second Problem.

I have divided ten into two portions: I have multiplied each of the parts by itself, and afterwards ten by itself: the product of ten by itself is equal to one of the two parts multiplied by itself, and afterwards by two and seven-ninths; or equal to the other multiplied by itself, and afterwards by six and one-fourth.*

Computation: Suppose one of the parts to be thing, and the other ten minus thing. You multiply thing by itself, it is a square; then by two and seven-ninths, this makes it two squares and seven-ninths of a square. You afterwards multiply ten by ten; it is a hundred, which must be equal to two squares and seven-ninths of a square. Reduce it to one square, through division by nine twenty-fifths;* this being its fifth and four-fifths of its fifth, take now also the fifth and four-fifths of the fifth of a hundred; this is thirty-six, which is equal to one square. Take its root, it is six. This is one of the two portions; and accordingly the other is four. This question leads you, therefore, to one of the six cases, namely, "squares equal to numbers."

Third Problem.

I have divided ten into two parts. I have afterwards divided the one by the other, and the quotient was four.*

Computation: Suppose one of the two parts to be thing, the other ten minus thing.

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* $(10 - x)x = \text{product}; x^2 = 4(10 - x)x = 40x - 5x^2; 5x^2 = 40x; x^2 = 8x; x = 8; (10 - x) = 2.$

* $100 = x^2 \times 2\frac{7}{9}; \frac{1}{2\frac{7}{9}} \times 100 = x^2; x = 6; 10 - x = 4.$

* $2\frac{7}{9} = \frac{25}{9}; \frac{1}{\frac{25}{9}} \times 100 = \frac{9}{25} \times 100 = 36; x^2 = 36; x = 6.$

* $\frac{10-x}{x} = 4; 10 = 5x; x = 2; (10 - x) = 8.$

Then you divide ten minus thing by thing, in order that four may be obtained. You know that if you multiply the quotient by the divisor, the sum which was divided is restored. In the present question the quotient is four and the divisor is thing. Multiply, therefore, four by thing; the result is four things, which are equal to the sum to be divided, which was ten minus thing. You now reduce it by thing, which you add to the four things. Then we have five things equal to ten; therefore one thing is equal to two, and this is one of the two portions. This question refers you to one of the six cases, namely, "roots equal to numbers."

Fourth Problem.

I have multiplied one-third of thing and one dirhem by one-fourth of thing and one dirhem, and the product was twenty.[*]

Computation: You multiply one-third of thing by one-fourth of thing; it is one-half of a sixth of a square. Further, you multiply one dirhem by one-third of thing, it is one-third of thing; and one dirhem by one-fourth of thing, it is one-fourth of thing; and one dirhem by one dirhem, it is one dirhem. The result of this is: the moiety of one-sixth of a square, and one-third of thing, and one-fourth of thing, and one dirhem, is equal to twenty dirhems. Subtract now the one dirhem from these twenty dirhems; the remainder is nineteen dirhems, equal to the moiety of one-sixth of a square, and one-third of thing and one-fourth of thing. Now make your square a whole one: you do this by multiplying all that you have by twelve. Thus you have one square and seven roots equal to two hundred and twenty-eight. Halve the number of the roots, and multiply it by itself; it is twelve and one-fourth. Add this to the number, that is, to two hundred and twenty-eight; the sum is two hundred and forty and one quarter. Extract the root of this; it is fifteen and a half. Subtract from this the moiety of the roots, that is, seven and a half; there remains eight, which is the square required.

Fifth Problem.

I have divided ten into two parts; and having divided the one by the other, and multiplied each of them by itself, and also I had added the products, the sum was fifty-eight dirhems.[*]

Computation: Suppose one of the two parts to be thing, and the other ten minus thing. Multiply ten minus thing by itself; it is a hundred and a square minus twenty things. Then multiply thing by thing; it is a square. Add both together. The sum is a hundred, plus two squares minus twenty things, which are equal to fifty-eight dirhems. Take now the twenty negative things from the hundred and the two squares, and add them to fifty-eight; then a hundred, plus two squares, are equal to fifty-eight dirhems and twenty things. Reduce this to one square, by taking the moiety of all you have. It is then: fifty dirhems and a square, which are equal to twenty-nine dirhems and ten things. Then reduce this, by taking twenty-nine from fifty: there remains twenty-one and a square, equal to ten things. Halve the number of the roots, it is five; multiply this by itself, it is twenty-five; take from this the twenty-one which are connected with the square, the remainder is four. Extract the root, it is two. Subtract this from the moiety of the roots, namely, from five, there remains three. This is one of the portions; the other is seven. This question refers you to one of the six cases, namely "squares and numbers equal to roots."

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[*] $(\frac{1}{3}x+1)(\frac{1}{4}x+1)=20$; $\frac{1}{12}x^2+\frac{7}{12}x+1=20$; $x^2+7x=228$.

[*] $(10-x)^2+x^2=58$; $100-20x+x^2+x^2=58$; $2x^2-20x+100=58$; $x^2-10x+50=29$; $x=5\pm\sqrt{25-29}=5\pm2=7$ or 3.

Sixth Problem.

I have multiplied one-third of a root by one-fourth of a root, and the product is equal to the root and twenty-four dirhems.*

Computation: Call the root thing; then one-third of thing is multiplied by one-fourth of thing; this is the moiety of one-sixth of the square, and is equal to thing and twenty-four dirhems. Multiply this moiety of one-sixth of the square by twelve, in order to make your square a whole one, and multiply also the thing by twelve, which yields twelve things; and also four-and-twenty by twelve: the product of the whole will be two hundred and eighty-eight dirhems and twelve roots, which are equal to one square. The moiety of the roots is six. Multiply this by itself, and add it to two hundred and eighty-eight, it will be three hundred and twenty-four. Extract the root from this, it is eighteen; add this to the moiety of the roots, which was six; the sum is twenty-four, and this is the square sought for.

This question refers you to one of the six cases, namely, “roots and numbers equal to squares.”

* $\frac{1}{3}x \times \frac{1}{4}x = x + 24$; $\frac{1}{12}x^2 = x + 24$; $x^2 = 12x + 288$; $x = 6 + \sqrt{36 + 288} = 6 + 18 = 24$.

Various Questions

If a person puts such a question to you as: "I have divided ten into two parts, and multiplying one of these by the other, the result was twenty-one;"* then you know that one of the two parts is thing, and the other ten minus thing. Multiply, therefore, thing by ten minus thing; then you have ten things minus a square, which is equal to twenty-one. Separate the square from the ten things, and add it to the twenty-one. Then you have ten things, which are equal to twenty-one dirhems and a square. Take away the moiety of the roots, and multiply the remaining five by itself; it is twenty-five. Subtract from this the twenty-one which are connected with the square; the remainder is four. Extract its root, it is two. Subtract this from the moiety of the roots, namely, five: there remain three, which is one of the two parts. Or, if you please, you may add the root of four to the moiety of the roots; the sum is seven, which is likewise one of the parts.

This is one of the problems which may be resolved by addition and subtraction.

If the question be: "I have divided ten into two parts, and having multiplied each part by itself, I have subtracted the smaller from the greater, and the remainder was forty;"* then the computation is—you multiply ten minus thing by itself, it is a hundred plus one square minus twenty things; and you also multiply thing by thing, it is one square. Subtract this from a hundred and a square minus twenty things, and you have a hundred, minus twenty things, equal to forty dirhems. Separate now the twenty things from a hundred, and add them to the forty; then you have a hundred, equal to twenty things and forty dirhems. Subtract now forty from a hundred; there remains sixty dirhems, equal to twenty things: therefore one thing is equal to three, which is one of the two parts.

If the question be: "I have divided ten into two parts, and having multiplied each part by itself, I have put them together, and have added to them the difference of the two parts previously to their multiplication, and the amount of all this is fifty-four;"* then the computation is this: You multiply ten minus thing by itself; it is a hundred and a square minus twenty things. Then multiply also the other thing of the ten by itself; it is one square. Add this together, it will be a hundred plus two squares minus twenty things. It was stated that the difference of the two parts before multiplication should be added to them. You say, therefore, the difference between them is ten minus two things. The result is a hundred and ten and two squares minus twenty-two things, which are equal to fifty-four dirhems. Having reduced and equalized this, you may say, a hundred and ten dirhems and two squares are equal to fifty-four dirhems and twenty-two things. Reduce now the two squares to one square, by taking the moiety of all you have. Thus it becomes fifty-five dirhems and a square, equal to twenty-seven dirhems and eleven things. Subtract twenty-seven from fifty-five, there remain twenty-eight dirhems and a square, equal to eleven things. Halve now the things, it will be five and a half; multiply this by itself, it is thirty and a quarter. Subtract from it the twenty-eight which are combined with the square, the remainder is two and a fourth. Extract its root, it is one and a half. Subtract this from the moiety of the roots, there remain four, which is one of the two parts.

If one say, "I have divided ten into two parts; and have divided the first by the second,

*$(10-x)x = 21$; $10x - x^2 = 21$; which is to be reduced to $x = 5 \pm \sqrt{25-21} = 5 \pm 2$.

*$(10-x)^2 - x^2 = 40$; $100 - 20x + x^2 - x^2 = 40$; $100 - 20x = 40$; $100 = 20x + 40$; $60 = 20x$; $x = 3$.

*$100 - 20x + x^2 + x^2 + 10 - 2x = 54$; $100 - 22x + 2x^2 = 54$; $55 - 11x + x^2 = 27$; $x^2 + 28 = 11x$; $x = \frac{11}{2} \pm \sqrt{\frac{121}{4} - 28} = \frac{11}{2} \pm \frac{3}{2} = 7$ or 4.

and the second by the first, and the sum of the quotient is two dirhems and one-sixth;"* then the computation is this: If you multiply each part by itself, and add the products together, then their sum is equal to one of the parts multiplied by the other, and again by the quotient which is two and one-sixth. Multiply, therefore, ten less thing by itself; it is a hundred and a square less twenty things. Multiply thing by thing; it is one square. Add this together; the sum is a hundred plus two squares less twenty things, which is equal to thing multiplied by ten less thing; that is, to ten things less a square, multiplied by the sum of the quotients arising from the division of the two parts, namely, two and one-sixth. We have, therefore, twenty-one things and two-thirds of thing less two squares and one-sixth, equal to a hundred plus two squares less twenty things. Reduce this by adding the two squares and one-sixth to a hundred plus two squares less twenty things, and add the twenty negative things from the hundred plus the two squares to the twenty-one things and two-thirds of thing. Then you have a hundred plus four squares and one-sixth of a square, equal to forty-one things and two-thirds of thing. Now reduce this to one square. You know that one square is obtained from four squares and one-sixth, by taking a fifth and one-fifth of a fifth.* Take, therefore, the fifth and one-fifth of a fifth of all that you have. Then it is twenty-four and a square, equal to ten roots; because ten is one-fifth and one-fifth of the fifth of the forty-one things and two-thirds of a thing. Now halve the roots; it gives five. Multiply this by itself; it is five-and-twenty. Subtract from this the twenty-four, which are connected with the square; the remainder is one. Extract its root; it is one. Subtract this from the moiety of the roots, which is five. There remains four, which is one of the two parts.

Observe that, in every case, where any two quantities whatsoever are divided, the first by the second and the second by the first, if you multiply the quotient of the one division by that of the other, the product is always one.*

If some one say: "You divide ten into two parts; multiply one of the two parts by five, and divide it by the other: then take the moiety of the quotient, and add this to the product of the one part, multiplied by five; the sum is fifty dirhems;"* then the computation is this: Take thing, and multiply it by five. This is now to be divided by the remainder of the ten, that is, by ten less thing; and of the quotient the moiety is to be taken.

You know that if you divide five things by ten less thing, and take the moiety of the quotient, the result is the same as if you divide the moiety of five things by ten less thing. Take, therefore, the moiety of five things; it is two things and a half: and this you require to divide by ten less thing. Now these two things and a half, divided by ten less thing, give a quotient which is equal to fifty less five things: for the question states: add this (the quotient) to the one part multiplied by five, the sum will be fifty. You have already observed, that if the quotient, or the result of the division, be multiplied by the divisor, the dividend, or capital to be divided, is restored. Now, your capital, in the present instance, is two things and a half. Multiply, therefore, ten less thing by fifty less five things. Then you have five hundred dirhems and five squares less a hundred things, which are equal to two things and a half. Reduce this to one square. Then it becomes a hundred dirhems and a square less twenty things, equal to the moiety of thing. Separate now the twenty things from the hundred dirhems and square, and add them to the half thing. Then you have a hundred dirhems and a square, equal to twenty things and a half. Now halve the

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* $100+2x^2-20x = x(10-x)\times 2\frac{1}{6} = 21\frac{2}{3}x - 2\frac{1}{6}x^2$; $100+4\frac{1}{6}x^2 = 41\frac{2}{3}x$; $24+x^2 = 10x$; $x = 5-\sqrt{25-24} = 5-1 = 4$ or 6.

* $4\frac{1}{6} = \frac{25}{6}$; $\frac{1}{4\frac{1}{6}} = \frac{6}{25} = \frac{1}{5} + \frac{1}{5}\frac{1}{5}$.

* $\frac{a}{b} \times \frac{b}{a} = 1$.

* $x^2 + 100 = 50x$; $x = 25 - \sqrt{625-100} = 25 - \sqrt{525}$.

things, multiply the moiety by itself, subtract from this the hundred, extract the root of the remainder, and subtract this from the moiety of the roots, which is ten and one-fourth: the remainder is eight; and this is one of the portions.

If some one say: "You divide ten into two parts: multiply the one by itself; it will be equal to the other taken eighty-one times."* Computation: You say, ten less thing, multiplied by itself, is a hundred plus a square less twenty things, and this is equal to eighty-one things. Separate the twenty things from a hundred and a square, and add them to eighty-one. It will then be a hundred plus a square, which is equal to a hundred and one roots. Halve the roots; the moiety is fifty and a half. Multiply this by itself, it is two thousand five hundred and fifty and a quarter. Subtract from this one hundred; the remainder is two thousand four hundred and fifty and a quarter. Extract the root from this; it is forty-nine and a half. Subtract this from the moiety of the roots, which is fifty and a half. There remains one, and this is one of the two parts.

If some one say: "I have purchased two measures of wheat or barley, each of them at a certain price. I afterwards added the expenses, and the sum was equal to the difference of the two prices, added to the difference of the measures."* Computation: Take what numbers you please, for it is indifferent; for instance, four and six. Then you say: I have bought each measure of the four for thing; and accordingly you multiply four by thing, which gives four things; and I have bought the six, each for the moiety of thing, for which I have bought the four; or, if you please, for one-third, or one-fourth, or for any other quota of that price, for it is indifferent. Suppose that you have bought the six measures for the moiety of thing, then you multiply the moiety of thing by six; this gives three things. Add them to the four things; the sum is seven things, which must be equal to the difference of the two quantities, which is two measures, plus the difference of the two prices, which is a moiety of thing. You have, therefore, seven things, equal to two and a moiety of thing. Remove, now, this moiety of thing, by subtracting it from the seven things. There remain six things and a half, equal to two dirhems: consequently, one thing is equal to four-thirteenths of a dirhem. The six measures were bought, each at one-half of thing; that is, at two-thirteenths of a dirhem. Accordingly, the expenses amount to eight-and-twenty thirteenths of a dirhem, and this sum is equal to the difference of the two quantities; namely, the two measures, the arithmetical equivalent for which is six-and-twenty thirteenths, added to the difference of the two prices, which is two-thirteenths: both differences together being likewise equal to twenty-eight parts.

If he say: "There are two numbers,* the difference of which is two dirhems. I have divided the smaller by the larger, and the quotient was just half a dirhem."* Suppose one of the two numbers to be thing, and the other to be thing plus two dirhems. By the division of thing by thing plus two dirhems, half a dirhem appears as quotient. You have already observed, that by multiplying the quotient by the divisor, the capital which you divided is restored. This capital, in the present case, is thing. Multiply, therefore, thing and two dirhems by half a dirhem, which is the quotient; the product is half one thing plus one dirhem; this is equal to thing. Remove, now, half a thing on account of the other

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*$100 - 20x + x^2 = 81x$; $x^2 + 100 = 101x$; $x = \frac{101}{2} - \sqrt{\frac{10201}{4} - 100} = 50\frac{1}{2} - 49\frac{1}{2} = 1$.

*The purchaser does not make a clear enunciation of the terms of his bargain. He intends to say, "I bought m bushels of wheat, and n bushels of barley, and the wheat was r times dearer than the barley. The sum I expended was equal to the difference in the quantities, added to the difference in the prices of the grain." If x is the price of the barley, rx is the price of the wheat; whence, $mrx + nx = (m-n) + (rx - x)$; $x = \frac{m-n}{mr+n-r+1}$.

*"*Squares*" in the original. The word square is used in the text to signify either, 1st, a square, properly so called, fractional or integral; 2d, a rational integer, not being a square number; 3d, a rational fraction, not being a square; 4th, a quadratic surd, fractional or integral.

*$\frac{x}{x+2} = \frac{1}{2}$; $\frac{1}{2}(x+2) = x$; $\frac{1}{2}x + 1 = x$; $1 = \frac{1}{2}x$; $x = 2$; the other is 4.

half thing; there remains one dirhem, equal to half a thing. Double it, then you have one thing, equal to two dirhems. Consequently, the other* number is four.

If some one say: "I have divided ten into two parts; I have multiplied the one by ten and the other by itself, and the products were the same;"* then the computation is this: You multiply thing by ten; it is ten things. Then multiply ten less thing by itself; it is a hundred and a square less twenty things, which is equal to ten things. Reduce this according to the rules, which I have above explained to you.

In like manner, if he say: "I have divided ten into two parts; I have multiplied one of the two by the other, and have then divided the product by the difference of the two parts before their multiplication, and the result of this division is five and one-fourth;"* the computation will be this: You subtract thing from ten; there remain ten less thing. Multiply the one by the other, it is ten things less a square. This is the product of the multiplication of one of the two parts by the other. At present you divide this by the difference between the two parts, which is ten less two things. The quotient of this division is, according to the statement, five and a fourth. If, therefore, you multiply five and one-fourth by ten less two things, the product must be equal to the above amount, obtained by multiplication, namely, ten things less one square. Multiply now five and one-fourth by ten less two squares. The result is fifty-two dirhems and a half less ten roots and a half, which is equal to ten roots less a square. Separate now the ten roots and a half from the fifty-two dirhems, and add them to the ten roots less a square; at the same time separate this square from them, and add it to the fifty-two dirhems and a half. Thus you find twenty roots and a half, equal to fifty-two dirhems and a half and one square. Now continue reducing it, conformably to the rules explained at the commencement of this book.

If the question be: "There is a square, two-thirds of one-fifth of which are equal to one-seventh of its root;"* then the square is equal to one root and half a seventh of a root; and the root consists of fourteen-fifteenths of the square. The computation is this: You multiply two-thirds of one-fifth of the square by seven and a half, in order that the square may be completed. Multiply that which you have already, namely, one-seventh of its root, by the same. The result will be, that the square is equal to one root and half a seventh of the root; and the root of the square is one and a half seventh; and the square is one and twenty-nine one hundred and ninety-sixths of a dirhem. Two-thirds of the fifth of this are thirty parts of the hundred and ninety-six parts. One-seventh of its root is likewise thirty parts of a hundred and ninety-six.

If the instance be: "Three-fourths of the fifth of a square are equal to four-fifths of its root,"* then the computation is this; You add one-fifth to the four-fifths, in order to complete the root. This is then equal to three and three-fourths of twenty parts, that is, to fifteen eightieths of the square. Divide now eighty by fifteen; the quotient is five and one-third. This is the root of the square, and the square is twenty-eight and four-ninths.

If some one say: "What is the amount of a square-root,* which, when multiplied by four times itself, amounts to twenty?"* the answer is this: If you multiply it by itself it will be five: it is therefore the root of five.

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* "*Square*" in the original.

* $10x = (10 - x)^2 = 100 - 20x + x^2$.

* $\frac{(10-x)x}{10-2x} = 5\frac{1}{4}$; $20\frac{1}{4}x = x^2 + 5\frac{1}{4}(10 - 2x)$; $x = 10 - 7\frac{1}{2} = 2\frac{1}{2}$.

* $\frac{2}{3} \cdot \frac{1}{5}x^2 = \frac{1}{7}x$; $\frac{2}{15}x^2 = \frac{1}{7}x$; $x^2 = \frac{15}{14}x$; $x = \frac{15}{14}$.

* $\frac{3}{4} \cdot \frac{1}{5}x^2 = \frac{4}{5}x$; $\frac{3}{20}x^2 = \frac{4}{5}x$; $\frac{3}{20}x = \frac{4}{5}$; $x = \frac{16}{3} = 5\frac{1}{3}$.

* "*Square*" in the original.

* $4x^2 = 20$; $x^2 = 5$; $x = \sqrt{5}$.

If somebody ask you for the amount of a square-root* which when multiplied by its third amounts to ten,* the solution is, that when multiplied by itself it will amount to thirty; and it is consequently the root of thirty.

If the question be: "To find a quantity,* which when multiplied by four times itself, gives one-third of the first quantity as product,"* the solution is, that if you multiply it by twelve times itself, the quantity itself must re-appear: it is the moiety of one moiety of one-third.

If the question be: "A square, which when multiplied by its root gives three times the original square as product,"* then the solution is: that if you multiply the root by one-third of the square, the original square is restored; its root must consequently be three, and the square itself nine.

If the instance be: "To find a square, four roots of which, multiplied by three roots, restore the square with a surplus of forty-four dirhems,"* then the solution is: that you multiply four roots by three roots, which gives twelve squares, equal to a square and forty-four dirhems. Remove now one square of the twelve on account of the one square connected with the forty-four dirhems. There remain eleven squares, equal to forty-four dirhems. Make the division, the result will be four, and this is the square.

If the instance be: "A square, four of the roots of which multiplied by five of its roots produce twice the square, with a surplus of thirty-six dirhems;"* then the solution is: that you multiply four roots by five roots, which gives twenty squares, equal to two squares and thirty-six dirhems. Remove two squares from the twenty on account of the other two. The remainder is eighteen squares, equal to thirty-six dirhems. Divide now thirty-six dirhems by eighteen; the quotient is two, and this is the square.

In the same manner, if the question be: "A square, multiply its root by four of its roots, and the product will be three times the square, with a surplus of fifty dirhems."* Computation: You multiply the root by four roots, it is four squares, which are equal to three squares and fifty dirhems. Remove three squares from the four; there remains one square, equal to fifty dirhems. One root of fifty, multiplied by four roots of the same, gives two hundred, which is equal to three times the square, and a residue of fifty dirhems.

If the instance be: "A square, which when added to twenty dirhems, is equal to twelve of its roots,"* then the solution is this: You say, one square and twenty dirhems are equal to twelve roots. Halve the roots and multiply them by themselves; this gives thirty-six. Subtract from this the twenty dirhems, extract the root from the remainder, and subtract it from the moiety of the roots, which is six. The remainder is the root of the square: it is two dirhems, and the square is four.

If the instance be: "To find a square, of which if one-third be added to three dirhems, and the sum be subtracted from the square, the remainder multiplied by itself restores the square;"* then the computation is this: If you subtract one-third and three dirhems from the square, there remain two-thirds of it less three dirhems. This is the root. Multiply therefore two-thirds of thing less three dirhems by itself. You say two-thirds by

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*"*Square*" in the original.

*$x \times \frac{x}{3} = 10$; $x^2 = 30$; $x = \sqrt{30}$.

*"*Square-root*" in the original.

*$x \times 4x = \frac{1}{3}x$; $4x^2 = \frac{1}{3}x$; $12x^2 = x$; $x = \frac{1}{12}$.

*$x^2 \times x = 3x^2$; $x^3 = 3x^2$; $x = 3$.

*$4x \times 3x = x^2 + 44$; $12x^2 = x^2 + 44$; $11x^2 = 44$; $x^2 = 4$; $x = 2$.

*$4x \times 5x = 2x^2 + 36$; $20x^2 = 2x^2 + 36$; $18x^2 = 36$; $x^2 = 2$.

*$x \times 4x = 3x^2 + 50$; $4x^2 = 3x^2 + 50$; $x^2 = 50$; $x = \sqrt{50}$.

*$x^2 + 20 = 12x$; $x = 6 \pm \sqrt{36 - 20} = 6 \pm 4 = 10$ or 2.

*$(x^2 - \frac{1}{3}x^2 - 3)^2 = x^2$; $\frac{2}{3}x^2 - 3 = x$; $\frac{4}{9}x^4 - 4x^2 + 9 = x^2$; $x^2 + 9 = 5x$; $x = 9$ or $2\frac{1}{4}$.

two-thirds is four ninths of a square; and less two-thirds by three dirhems is two roots; and again, two-thirds by three dirhems is two roots; and less three dirhems by less three dirhems is nine dirhems. You have, therefore, four-ninths of a square and nine dirhems less four roots, which are equal to one root. Add the four roots to the one root, then you have five roots, which are equal to four-ninths of a square and nine dirhems. Complete now your square; that is, multiply the four-ninths of a square by two and a fourth, which gives one square; multiply likewise the nine dirhems by two and a quarter; this gives twenty and a quarter; finally, multiply the five roots by two and a quarter; this gives eleven roots and a quarter. You have, therefore, a square and twenty dirhems and a quarter, equal to eleven roots and a quarter. Reduce this according to what I taught you about halving the roots.

If the instance be: "To find a number,[*] one-third of which, when multiplied by one-fourth of it, restores the number,"[*] then the computation is: You multiply one-third of thing by one-fourth of thing, this gives one-twelfth of a square, equal to thing, and the square is equal to twelve things, which is the root of one hundred and forty-four.

If the instance be: "A number,[*] one-third of which and one dirhem multiplied by one-fourth of it and two dirhems restore the number, with a surplus of thirteen dirhems;"[*] then the computation is this: You multiply one-third of thing by one-fourth of thing, this gives half one-sixth of a square; and you multiply two dirhems by one-third of thing, this gives two-thirds of a root; and one dirhem by one-fourth of thing gives one-fourth of a root; and one dirhem by two dirhems gives two dirhems. This altogether is one-twelfth of a square and two dirhems and eleven-twelfths of a thing, equal to thing and thirteen dirhems. Remove now two dirhems from thirteen, on account of the other two dirhems, the remainder is eleven dirhems. Remove then the eleven-twelfths of a root from the one (root on the opposite side), there remains one-twelfth of a root and eleven dirhems, equal to one-twelfth of a square. Complete the square: that is, multiply it by twelve, and do the same with all you have. The product is a square, which is equal to a hundred and thirty-two dirhems and one root. Reduce this, according to what I have taught you, it will be right.

If the instance be: "A dirhem and a half to be divided among one person and certain persons, so that the share of the one person be twice as many dirhems as there are other persons;"[*] then the Computation is this: You say, the one person and some persons are one and thing: it is the same as if the question had been one dirhem and a half to be divided by one and thing, and the share of one person to be equal to two things. Multiply, therefore, two things by one and thing; it is two squares and two things, equal to one dirhem and a half. Reduce them to one square: that is, take the moiety of all you have. You say, therefore, one square and one thing are equal to three-fourths of a dirhem. Reduce this, according to what I have taught you in the beginning of this work.

If the instance be: "A number,[*] you remove one-third of it, and one-fourth of it, and four dirhems: then you multiply the remainder by itself, and the number,[*] is restored,

[*] "*Square*" in the original.

[*] $\frac{1}{3}x \times \frac{1}{4}x = x$; $\frac{1}{12}x^2 = x$; $x^2 = 12x$; $x = 12$.

[*] "*Square*" in the original.

[*] $\frac{x}{3} + 1 \times \frac{x}{4} + 2 = x + 13$; $\frac{1}{12}x^2 + \frac{11}{12}x + 2 = x + 13$; $\frac{1}{12}x^2 + 11 = \frac{1}{12}x + 13$; $x^2 = 132 + x$; $x = 12$.

[*] The enunciation in the original is faulty, and I have altered it to correspond with the computation. But in the computation, x, the number of persons, is fractional! I am unable to correct the passage satisfactorily. $x^2 + x = \frac{3}{2}$; $x = \sqrt{1\frac{1}{4} + \frac{1}{2}} - \frac{1}{2}$.

[*] "*Square*" in the original.

[*] "*Square*" in the original.

with a surplus of twelve dirhems;”* then the computation is this: You take thing, and subtract from it one-third and one-fourth; there remain five-twelfths of thing. Subtract from this four dirhems; the remainder is five-twelfths of thing less four dirhems. Multiply this by itself. Thus the five parts become five-and-twenty parts; and if you multiply twelve by itself, it is a hundred and forty-four. This makes, therefore, five and twenty hundred and forty-fourths of a square. Multiply then the four dirhems twice by the five-twelfths; this gives forty parts, every twelve of which make one root (forty-twelfths); finally, the four dirhems, multiplied by four dirhems, give sixteen dirhems to be added. The forty-twelfths are equal to three roots and one-third of a root, to be subtracted. The whole product is, therefore, twenty-five-hundred-and-forty-fourths of a square and sixteen dirhems less three roots and one-third of a root, equal to the original number,* which is thing and twelve dirhems. Reduce this, by adding the three roots and one-third to the thing and twelve dirhems. Thus you have four roots and one-third of a root and twelve dirhems. Go on balancing, and subtract the twelve (dirhems) from sixteen; there remain four dirhems and five-and-twenty-hundred-and-forty-fourths of a square, equal to four roots and one-third. Now it is necessary to complete the square. This you can accomplish by multiplying all you have by five and nineteen twenty-fifths. Multiply, therefore, the twenty-five-one-hundred-and-forty-fourths of a square by five and nineteen twenty-fifths. This gives a square. Then multiply the four (dirhems) by five and nineteen twenty-fifths; this gives twenty-three dirhems and one twenty-fifth. Then multiply four roots and one-third by five and nineteen twenty-fifths; this gives twenty-four roots and twenty-four twenty-fifths of a root. Now halve the number of the roots: the moiety is twelve roots and twelve twenty-fifths of a root. Multiply this by itself. It is one hundred-and-fifty-five dirhems and four-hundred-and-sixty-nine six-hundred-and-twenty-fifths. Subtract from this the twenty-three dirhems and the one twenty-fifth connected with the square. The remainder is one-hundred-and-thirty-two and four-hundred-and-forty-six six-hundred-and-twenty-fifths. Take the root of this: it is eleven dirhems and thirteen twenty-fifths. Add this to the moiety of the roots, which was twelve dirhems and twelve twenty-fifths. The sum is twenty-four. It is the number which you sought. When you subtract its third and its fourth and four dirhems, and multiply the remainder by itself, the number is restored, with a surplus of twelve dirhems.

If the question be: “To find a square-root,* which, when multiplied by two-thirds of itself, amounts to five;”* then the computation is this: You multiply one thing by two-thirds of thing; the product is two-thirds of square, equal to five. Complete it by adding its moiety to it, and add to five likewise its moiety. Thus you have a square, equal to seven and a half. Take its root; it is the thing which you required, and which, when multiplied by two-thirds of itself, is equal to five.

If the instance be: “Two numbers,* the difference of which is two dirhems; you divide the small one by the great one, and the quotient is equal to half a dirhem;”* then the computation is this: Multiply thing and two dirhems by the quotient, that is a half. The product is half a thing and one dirhem, equal to thing. Remove now half a dirhem on account of the half dirhem on the other side. The remainder is one dirhem, equal to half

*$(x - \frac{1}{3}x - \frac{1}{4}x - 4)^2 = x + 12$; $\frac{5}{12}x - 4$; $x^2 + 23\frac{1}{4} = 24\frac{1}{4}x$.

*“*Square*” in the original.

*“*Square*” in the original.

*$x \times \frac{2}{3}x = 5$; $\frac{2}{3}x^2 = 5$; $x^2 = 7\frac{1}{2}$; $x = \sqrt{7\frac{1}{2}}$.

*“*Squares*” in the original.

*$\frac{x}{x+2} = \frac{1}{2}$; $\frac{1}{2}(x+2) = x$; $1 + \frac{1}{2}x = x$; $1 = \frac{1}{2}x$; $x = 2$; the other is 4.

a thing. Double it; then you have thing, equal to two dirhems. This is one of the two* numbers, and the other is four.

Instance: "You divide one dirhem amongst a certain number of men, which number is thing. Now you add one man more to them, and divide again one dirhem amongst them; the quota of each is then one-sixth of a dirhem less than at the first time."* Computation: You multiply the first number of men, which is thing, by the difference of the share for each of the other number. Then multiply the product by the first and second number of men, and divide the product by the difference of these two numbers. Thus you obtain the sum which shall be divided. Multiply, therefore, the first number of men, which is thing, by the one-sixth, which is the difference of the shares; this gives one-sixth of root. Then multiply this by the original number of the men, and that of the additional one, that is to say, by thing plus one. The product is one-sixth of square and one-sixth of root divided by one dirhem, and this is equal to one dirhem. Complete the square which you have through multiplying it, by six. Then you have a square and a root equal to six dirhems. Halve the root and multiply the moiety by itself, it is one-fourth. Add this to the six; take the root of the sum and subtract from it the moiety of the root, which you have multiplied by itself, namely, a half. The remainder is the first number of men; which in this instance is two.

If the instance be: "To find a square-root,* which when multiplied by two-thirds of itself amounts to five:"* then the computation is this: If you multiply it by itself, it gives seven and a half. Say, therefore, it is the root of seven and a half multiplied by two-thirds of the root of seven and a half. Multiply then two-thirds by two-thirds, it is four-ninths; and four-ninths multiplied by seven and a half is three and a third. The root of three and a third is two-thirds of the root of seven and a half. Multiply three and a third by seven and a half; the product is twenty-five, and its root is five.

If the instance be: "A square multiplied by three of its roots is equal to five times the original square;"* then this is the same as if it had been said, a square, which when multiplied by its root, is equal to the first square and two-thirds of it. Then the root of the square is one and two-thirds, and the square is two dirhems and seven-ninths.

If the instance be: "Remove one-third from a square, then multiply the remainder by three roots of the first square, and the first square will be restored."* Computation: If you multiply the first square, before removing two-thirds from it, by three roots of the same, then it is one square and a half; for according to the statement two-thirds of it multiplied by three roots give one square; and, consequently, the whole of it multiplied by three roots of it gives one square and a half. This entire square, when multiplied by one root, gives half a square; the root of the square must therefore be a half, the square one-fourth, two-thirds of the square one-sixth, and three roots of the square one and a half. If you multiply one-sixth by one and a half, the product is one-fourth, which is the square.

Instance: "A square; you subtract four roots of the same, then take one-third of the remainder; this is equal to the four roots." The square is two hundred and fifty-six.* Com-

* "*Squares*" in the original.

* $\frac{1}{x} - \frac{1}{x+1} = \frac{1}{6}$; $\frac{x+1-x}{x(x+1)} = \frac{1}{6}$; $\frac{1}{x(x+1)} = \frac{1}{6}$; $x^2 + x = 6$; $x = \sqrt{\frac{1}{4} + 6} - \frac{1}{2} = 2$.

* "*Square*" in the original.

* $\frac{2}{3}x^2 = 5$; $x^2 = 7\frac{1}{2}$; $x = \sqrt{7\frac{1}{2}}$.

* $x^2 \times 3x = 5x^2$; $3x^3 = 5x^2$; $3x = 5$; $x = 1\frac{2}{3}$; $x^2 = 2\frac{7}{9}$.

* $(x^2 - \frac{1}{3}x^2) \times 3x = x^2$; $\frac{2}{3}x^2 \times 3x = x^2$; $2x^3 = x^2$; $x = \frac{1}{2}$; $x^2 = \frac{1}{4}$.

* $x^2 - 4x = \frac{1}{3}(x^2 - 4x)$ gives $\frac{2}{3}(x^2 - 4x) = 4x$; $x^2 - 4x = 6x$; $x^2 = 10x$; $x = 10$; $x^2 = 100$. *[Note: There appears to be a discrepancy in the original text, which states 256.]*

putation: You know that one-third of the remainder is equal to four roots; consequently, the whole remainder must be twelve roots; add to this the four roots; the sum is sixteen, which is the root of the square.

Instance: "A square; you remove one root from it and if you add to this root a root of the remainder, the sum is two dirhems."* Then, this is the root of a square, which, when added to the root of the same square, less one root, is equal to two dirhems. Subtract from this one root of the square, and subtract also from the two dirhems one root of the square. Then two dirhems less one root multiplied by itself is four dirhems and one square less four roots, and this is equal to a square less one root. Reduce it, and you find a square and four dirhems, equal to a square and three roots. Remove square by square; there remain three roots, equal to four dirhems; consequently, one root is equal to one dirhem and one-third. This is the root of the square, and the square is one dirhem and seven ninths of a dirhem.

Instance: "Subtract three roots from a square, then multiply the remainder by itself, and the square is restored."* You know by this statement that the remainder must be a root likewise; and that the square consists of four such roots; consequently, it must be sixteen.

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* $\sqrt{x^2 - x} + x = 2$; $\sqrt{x^2 - x} = 2 - x$; $x^2 - x = 4 + x^2 - 4x$; $3x = 4$; $x = 1\frac{1}{3}$; $x^2 = 1\frac{7}{9}$.

* $x^2 - 3x = x$; $x^2 = 4x$; $x = 4$.

On Mercantile Transactions

You know that all mercantile transactions of people, such as buying and selling, exchange and hire, comprehend always two notions and four numbers, which are stated by the enquirer; namely, measure and price, and quantity and sum. The number which expresses the measure is inversely proportionate to the number which expresses the sum, and the number of the price inversely proportionate to that of the quantity. Three of these four numbers are always known, one is unknown, and this is implied when the person inquiring says *how much?* and it is the object of the question.

The computation in such instances is this, that you try the three given numbers; two of them must necessarily be inversely proportionate the one to the other. Then you multiply these two proportionate numbers by each other, and you divide the product by the third given number, the proportionate of which is unknown. The quotient of this division is the unknown number, which the inquirer asked for; and it is inversely proportionate to the divisor.*

Examples.

For the first case: If you are told, "ten for six, how much for four?" then ten is the measure; six is the price; the expression *how much* implies the unknown number of the quantity; and *four* is the number of the sum. The number of the measure, which is ten, is inversely proportionate to the number of the sum, namely, *four*. Multiply, therefore, ten by four, that is to say, the two known proportionate numbers by each other; the product is forty. Divide this by the other known number, which is that of the price, namely, six. The quotient is six and two-thirds; it is the unknown number, implied in the words of the question "how much?" it is the quantity, and inversely proportionate to the six, which is the price.

For the second case: Suppose that some one ask this question: "ten for eight, what must be the sum for four?" This is also sometimes expressed thus: "What must be the price of four of them?" Ten is the number of the measure, and is inversely proportionate to the unknown number of the sum, which is involved in the expression *how much* of the statement. Eight is the number of the price, and this is inversely proportionate to the known number of the quantity, namely, four. Multiply now the two known proportionate numbers one by the other, that is to say, four by eight. The product is thirty-two. Divide this by the other known number, which is that of the measure, namely, ten. The quotient is three and one-fifth; this is the number of the sum, and inversely proportionate to the ten which was the divisor. In this manner all computations in matters of business may be solved.

If somebody says, "a workman receives a pay of ten dirhems per month; how much must be his pay for six days?" Then you know that six days are one-fifth of the month; and that his portion of the dirhems must be proportionate to the portion of the month. You calculate it by observing that one month, or thirty days, is the measure, ten dirhems the price, six days the quantity, and his portion the sum. Multiply the price, that is, ten, by the quantity, which is proportionate to it, namely, six; the product is sixty. Divide this by thirty, which is the known number of the measure. The quotient is two dirhems, and this is the sum.

This is the proceeding by which all transactions concerning exchange or measures or weights are settled.

*If a is given for b, and A for B; then $a : b :: A : B$ or $aB = bA$; $\therefore a = \frac{bA}{B}$ and $b = \frac{aB}{A}$.

Mensuration

KNOW THAT the meaning of the expression "one by one" is mensuration: one yard (in length) by one yard (in breadth) being understood.

Every quadrangle of equal sides and angles, which has one yard for every side, has also one for its area. Has such a quadrangle two yards for its side, then the area of the quadrangle is four times the area of a quadrangle, the side of which is one yard. The same takes place with three by three, and so on, ascending or descending: for instance, a half by a half, which gives a quarter, or other fractions, always following the same rule.

A quadrate, every side of which is half a yard, is equal to one-fourth of the figure which has one yard for its side. In the same manner, one-third by one-third, or one-fourth by one-fourth, or one-fifth by one-fifth, or two-thirds by a half, or more or less than this, always according to the same rule.

One side of an equilateral quadrangular figure, taken once, is its root; or if the same be multiplied by two, then it is like two of its roots, whether it be small or great.

If you multiply the height of any equilateral triangle by the moiety of the basis upon which the line marking the height stands perpendicularly, the product gives the area of that triangle.

In every equilateral quadrangle, the product of one diameter multiplied by the moiety of the other will be equal to the area of it.

In any circle, the product of its diameter, multiplied by three and one-seventh, will be equal to the periphery. This is the rule generally followed in practical life, though it is not quite exact. The geometricians have two other methods. One of them is, that you multiply the diameter by itself; then by ten, and hereafter take the root of the product; the root will be the periphery. The other method is used by the astronomers among them: it is this, that you multiply the diameter by sixty-two thousand eight hundred and thirty-two and then divide the product by twenty thousand; the quotient is the periphery. Both methods come very nearly to the same effect.[*]

If you divide the periphery by three and one-seventh, the quotient is the diameter.

The area of any circle will be found by multiplying the moiety of the circumference by the moiety of the diameter; since, in every polygon of equal sides and angles, such as triangles, quadrangles, pentagons, and so on, the area is found by multiplying the moiety of the circumference by the moiety of the diameter of the middle circle that may be drawn through it.

If you multiply the diameter of any circle by itself, and subtract from the product one-seventh and half one-seventh of the same, then the remainder is equal to the area of the circle. This comes very nearly to the same result with the method given above.[*]

Every part of a circle may be compared to a bow. It must be either exactly equal to half the circumference, or less or greater than it. This may be ascertained by the arrow of the bow. When this becomes equal to the moiety of the chord, then the arc is exactly the moiety of the circumference: is it shorter than the moiety of the chord, then the bow is less than half the circumference; is the arrow longer than half the chord, then the bow comprises more than half the circumference.

If you want to ascertain the circle to which it belongs, multiply the moiety of the chord by itself, divide it by the arrow, and add the quotient to the arrow, the sum is the diameter of the circle to which this bow belongs.

[*]The three formulas are: 1^{st}, $3\frac{1}{7}d = p$ i.e. $3.1428\,d$; 2^d, $\sqrt{10d^2} = p$ i.e. $3.16227\,d$; 3^d, $\frac{d\times 62832}{20000} = p$ i.e. $3.1416\,d$.

[*]The area of a circle whose diameter is d is $\pi\frac{d^2}{4} = d^2 - \frac{d^2}{7} - \frac{1}{2}\cdot\frac{d^2}{7} = d^2(1-\frac{3}{14}) = \frac{11}{14}d^2 \approx 0.7857d^2$. The value $\frac{22}{7}\cdot\frac{1}{4} = \frac{11}{14} \approx 0.7857$.

If you want to compute the area of the bow, multiply the moiety of the diameter of the circle by the moiety of the bow, and keep the product in mind. Then subtract the arrow of the bow from the moiety of the diameter of the circle, if the bow is smaller than half the circle; or if it is greater than half the circle, subtract half the diameter of the circle from the arrow of the bow. Multiply the remainder by the moiety of the chord of the bow, and subtract the product from that which you have kept in mind if the bow is smaller than the moiety of the circle, or add it thereto if the bow is greater than half the circle. The sum after the addition, or the remainder after the subtraction, is the area of the bow.

The bulk of a quadrangular body will be found by multiplying the length by the breadth, and then by the height.

If it is of another shape than the quadrangular (for instance, circular or triangular), so, however, that a line representing its height may stand perpendicularly on its basis, and yet be parallel to the sides, you must calculate it by ascertaining at first the area of its basis. This, multiplied by the height, gives the bulk of the body.

Cones and pyramids, such as triangular or quadrangular ones, are computed by multiplying one-third of the area of the basis by the height.

On the Rectangular Triangle.

Observe, that in every rectangular triangle the two short sides, each multiplied by itself and the products added together, equal the product of the long side multiplied by itself.

The proof of this is the following. We draw a quadrangle, with equal sides and angles $ABCD$. We divide the line AC into two moieties in the point H, from which we draw a parallel to the point R. Then we divide, also, the line AB into two moieties at the point T, and draw a parallel to the point G. Then the quadrate $ABCD$ is divided into four quadrangles of equal sides and angles, and of equal area; namely, the squares AK, CK, BK, and DK. Now, we draw from the point H to the point T a line which divides the quadrangle AK into two equal parts: thus there arise two triangles from the quadrangle, namely, the triangles ATH and HKT. We know that AT is the moiety of AB, and that AH is equal to it, being the moiety of AC; and the line TH joins them opposite the right angle. In the same manner we draw lines from T to R, and from R to G, and from G to H. Thus from all the squares eight equal triangles arise, four of which must, consequently, be equal to the moiety of the great quadrate AD. We know that the line AT multiplied by itself is like the area of two triangles, and AK gives the area of two triangles equal to them; the sum of them is therefore four triangles. But the line HT, multiplied by itself, gives likewise the area of four such triangles. We perceive, therefore, that the sum of AT multiplied by itself, added to AH multiplied by itself, is equal to TH multiplied by itself.

This is the observation which we were desirous to elucidate.

Quadrangles are of five kinds: firstly, with right angles and equal sides; secondly, with right angles and unequal sides; thirdly, the rhombus, with equal sides and unequal angles; fourthly, the rhomboid, the length of which differs from its breadth, and the angles of which are unequal, only that the two long and the two short sides are respectively of equal length; fifthly, quadrangles with unequal sides and angles.

First kind.—The area of any quadrangle with equal sides and right angles, or with unequal sides and right angles, may be found by multiplying the length by the breadth. The product is the area. For instance: a quadrangular piece of ground, every side of which has five yards, has an area of five-and-twenty square yards.

Second kind.—A quadrangular piece of ground, the two long sides of which are of eight yards each, while the breadth is six. You find the area by multiplying six by eight, which yields forty-eight yards.

Third kind, the Rhombus.—Its sides are equal: let each of them be five, and let its diagonals be, the one eight and the other six yards. You may then compute the area, either from one of the diagonals, or from both. As you know them both, you multiply the one by the moiety of the other, the product is the area: that is to say, you multiply eight by three, or six by four; this yields twenty-four yards, which is the area.[*] If you know only one of the diagonals, then you are aware, that there are two triangles, two sides of each of which have every one five yards, while the third is the diagonal. Hereafter you can make the computation according to the rules for the triangles. This is the figure.

The fourth kind, or Rhomboid, is computed in the same way as the rhombus.

The other quadrangles are calculated by drawing a diagonal, and computing them as triangles.

Triangles are of three kinds, acute-angular, obtuse-angular, or rectangular. The peculiarity of the rectangular triangle is, that if you multiply each of its two short sides by itself, and then add them together, their sum will be equal to the long side multiplied by itself. The character of the acute-angled triangle is this: if you multiply every one of its two short sides by itself, and add the products, their sum is more than the long side alone multiplied by itself. The definition of the obtuse-angled triangle is this: if you multiply its two short sides each by itself, and then add the products, their sum is less than the product of the long side multiplied by itself.

The rectangular triangle has two cathetes and an hypotenuse. It may be considered as the moiety of a quadrangle. You find its area by multiplying one of its cathetes by the moiety of the other. The product is the area.

Examples.—A rectangular triangle; one cathete being six yards, the other eight, and the hypotenuse ten. You make the computation by multiplying six by four: this gives twenty-four, which is the area. Or if you prefer, you may also calculate it by the height, which rises perpendicularly from the longest side of it: for the two short sides may themselves be considered as two heights. If you prefer this, you multiply the height by the moiety of the basis. The product is the area.

Second kind.—An equilateral triangle with acute angles, every side of which is ten yards long. Its area may be ascertained by the line representing its height and the point from which it rises.[*] Observe, that in every isosceles triangle, a line to represent the height drawn to the basis rises from the latter in a right angle, and the point from which it proceeds is always situated in the midst of the basis; if, on the contrary, the two sides are not equal, then this point never lies in the middle of the basis. In the case now before us we perceive, that towards whatever side we may draw the line which is to represent the height, it must necessarily always fall in the middle of it, where the length of the basis is five. Now the height will be ascertained thus. You multiply five by itself; then multiply one of the sides, that is ten, by itself, which gives a hundred. Now you subtract from this the product of five multiplied by itself, which is twenty-five. The remainder is seventy-five, the root of which is the height. This is a line common to two rectangular triangles. If you want to find the area, multiply the root of seventy-five by the moiety of the basis, which is five. This you perform by multiplying at first five by itself; then you may say, that the root of seventy-five is to be multiplied by the root of twenty-five. Multiply seventy-five by twenty-five. The product is one thousand eight hundred and

[*] If the two diagonals are d and d', and the side s, the area of the rhombus is $\frac{dd'}{2} = d \times \sqrt{s^2 - \frac{d^2}{4}}$.

[*] The height of the equilateral triangle whose side is 10, is $\sqrt{10^2 - 5^2} = \sqrt{75}$, and the area of the triangle is $5\sqrt{75} = 25\sqrt{5}$. The root is $43.3+$.

seventy-five; take its root, it is the area: it is forty-three and a little.

There are also acute-angled triangles, with different sides. Their area will be found by means of the line representing the height and the point from which it proceeds. Take, for instance, a triangle, one side of which is fifteen yards, another fourteen, and the third thirteen yards.

In order to find the point from which the line marking the height does arise, you may take for the basis any side you choose; e.g. that which is fourteen yards long. The point from which the line representing the height does arise, lies in this basis at an unknown distance from either of the two other sides. Let us try to find its unknown distance from the side which is thirteen yards long. Multiply this distance by itself; it becomes an [unknown] square. Subtract this from thirteen multiplied by itself; that is, one hundred and sixty-nine. The remainder is one hundred and sixty-nine less a square. The root from this is the height. The remainder of the basis is fourteen less thing. We multiply this by itself; it becomes one hundred and ninety-six, and a square less twenty-eight things. We subtract this from fifteen multiplied by itself; the remainder is twenty-nine dirhems and twenty-eight things less one square. The root of this is the height. As, therefore, the root of this is the height, and the root of one hundred and sixty-nine less square is the height likewise, we know that they both are the same.* Reduce them, by removing square against square, since both are negatives. There remain twenty-nine [dirhems] plus twenty-eight things, which are equal to one hundred and sixty-nine. Subtract now twenty-nine from one hundred and sixty-nine. The remainder is one hundred and forty, equal to twenty-eight things. One thing is, consequently, five. This is the distance of the said point from the side of thirteen yards. The complement of the basis towards the other side is nine. Now in order to find the height, you multiply five by itself, and subtract it from the contiguous side, which is thirteen, multiplied by itself. The remainder is one hundred and forty-four. Its root is the height. It is twelve. The height forms always two right angles with the basis, and it is called the column, on account of its standing perpendicularly. Multiply the height into half the basis, which is seven. The product is eighty-four, which is the area.

The third species is that of the obtuse-angled triangle with one obtuse angle and sides of different length. For instance, one side being six, another five, and the third nine. The area of such a triangle will be found by means of the height and of the point from which a line representing the same arises. This point can, within such a triangle, lie only in its longest side. Take therefore this as the basis: for if you choose to take one of the short sides as the basis, then this point would fall beyond the triangle. You may find the distance of this point, and the height, in the same manner, which I have shown in the acute-angled triangle; the whole computation is the same.

We have above treated at length of the circles, of their qualities and their computation. The following is an example: If a circle has seven for its diameter, then it has twenty-two for its circumference. Its area you find in the following manner: Multiply the moiety of the diameter, which is three and a half, by the moiety of the circumference, which is eleven. The product is thirty-eight and a half, which is the area. Or you may also multiply the diameter, which is seven, by itself: this is forty-nine; subtracting herefrom one-seventh and half one-seventh, which is ten and a half, there remain thirty-eight and a half, which is the area.

If some one inquires about the bulk of a pyramidal pillar, its base being four yards by four yards, its height ten yards, and the dimensions at its upper extremity two yards by two yards; then we know already that every pyramid is decreasing towards its top,

* $\sqrt{169 - x^2} = \sqrt{29 + 28x - x^2}$; $169 = 29 + 28x$; $140 = 28x$; $x = 5$.

and that one-third of the area of its basis, multiplied by the height, gives its bulk. The present pyramid has no top. We must therefore seek to ascertain what is wanting in its height to complete the top. We observe, that the proportion of the entire height to the ten, which we have now before us, is equal to the proportion of four to two. Now as two is the moiety of four, ten must likewise be the moiety of the entire height, and the whole height of the pillar must be twenty yards. At present we take one-third of the area of the basis, that is, five and one-third, and multiply it by the length, which is twenty. The product is one hundred and six yards and two-thirds. Herefrom we must then subtract the piece, which we have added in order to complete the pyramid. This we perform by multiplying one and one-third, which is one-third of the product of two by two, by ten: this gives thirteen and a third. This is the piece which we have added in order to complete the pyramid. Subtracting this from one hundred and six yards and two-thirds, there remain ninety-three yards and one-third: and this is the bulk of the mutilated pyramid.

If the pillar has a circular basis, subtract one-seventh and half a seventh from the product of the diameter multiplied by itself, the remainder is the basis.

If some one says: "There is a triangular piece of land, two of its sides having ten yards each, and the basis twelve; what must be the length of one side of a quadrate situated within such a triangle?" the solution is this. At first you ascertain the height of the triangle, by multiplying the moiety of the basis, (which is six) by itself, and subtracting the product, which is thirty-six, from one of the two short sides multiplied by itself, which is one hundred; the remainder is sixty-four: take the root from this; it is eight. This is the height of the triangle. Its area is, therefore, forty-eight yards: such being the product of the height multiplied by the moiety of the basis, which is six. Now we assume that one side of the quadrate inquired for is thing. We multiply it by itself; thus it becomes a square, which we keep in mind. We know that there must remain two triangles on the two sides of the quadrate, and one above it. The two triangles on both sides of it are equal to each other: both having the same height and being rectangular. You find their area by multiplying thing by six less half a thing, which gives six things less half a square. This is the area of both the triangles on the two sides of the quadrate together. The area of the upper triangle will be found by multiplying eight less thing, which is the height, by half one thing. The product is four things less half a square. This altogether is equal to the area of the quadrate plus that of the three triangles: or, ten things are equal to forty-eight, which is the area of the great triangle. One thing from this is four yards and four-fifths of a yard; and this is the length of any side of the quadrate.

On Legacies

The problems in this chapter may be considered as belonging rather to Law than to Algebra, as they contain little more than enunciations of the law of inheritance in certain complicated cases.

On Capital, and Money lent.

"A man dies, leaving two sons behind him, and bequeathing one-third of his capital to a stranger. He leaves ten dirhems of property and a claim of ten dirhems upon one of the sons."

Computation: You call the sum which is taken out of the debt thing. Add this to the capital, which is ten dirhems. The sum is ten and thing. Subtract one-third of this, since he has bequeathed one-third of his property, that is, three dirhems and one-third of thing. The remainder is six dirhems and two-thirds of thing. Divide this between the two sons. The portion of each of them is three dirhems and one-third plus one-third of thing. This is equal to the thing which was sought for.*

Reduce it, by removing one-third from thing, on account of the other third of thing. There remain two-thirds of thing, equal to three dirhems and one-third. It is then only required that you complete the thing, by adding to it as much as one half of the same; accordingly, you add to three and one-third as much as one-half of them: This gives five dirhems, which is the thing that is taken out of the debts.

If he leaves two sons and ten dirhems of capital and a demand of ten dirhems against one of the sons, and bequeaths one-fifth of his property and one dirhem to a stranger, the computation is this: Call the sum which is taken out of the debt, thing. Add this to the property; the sum is thing and ten dirhems. Subtract one-fifth of this, since he has bequeathed one-fifth of his capital, that is, two dirhems and one-fifth of thing; the remainder is eight dirhems and four-fifths of thing. Subtract also the one dirhem which he has bequeathed; there remain seven dirhems and four-fifths of thing. Divide this between the two sons; there will be for each of them three dirhems and a half plus two-fifths of thing; and this is equal to one thing.*

Reduce it by subtracting two-fifths of thing from thing. Then you have three-fifths of thing, equal to three dirhems and a half. Complete the thing by adding to it two-thirds of the same: add as much to the three dirhems and a half; namely, two dirhems and one-third; the sum is five and five-sixths. This is the thing, or the amount which is taken from the debt.

If he leaves three sons, and bequeaths one-fifth of his property less one dirhem, leaving ten dirhems of capital and a demand of ten dirhems against one of the sons, the computation is this: You call the sum which is taken from the debt thing. Add this to the capital; it gives ten and thing. Subtract from this one-fifth of it for the legacy; it is two dirhems and one-fifth of thing. There remain eight dirhems and four-fifths of thing; add to this one dirhem, since he stated "less one dirhem." Thus you have nine dirhems and four-fifths of thing. Divide this between the three sons. There will be for each son three dirhems, and one-fifth and one-third and one-fifth of thing. This equals one thing.*

*If a father dies, leaving n sons, one of whom owes the father a sum exceeding an nth part of the residue of the father's estate, after paying legacies, then such son retains the whole sum which he owes the father: part, as a set-off against his share of the residue, the surplus as a gift from the father. In the present example, let each son's share of the residue be equal to x. Then $\frac{2}{3}[10+x]=2x; \therefore x=5$. The stranger receives 5; and the son, who is not indebted to the father, receives 5.

*$\frac{4}{5}[10+x]-1=2x; \therefore \frac{4}{5}[10+x]-1=2x; \frac{2}{3}[10+x]-1=x$; The stranger receives $\frac{1}{5}[10+x]+1$.

*The stranger receives $\frac{1}{5}[10+x]-1$.

Subtract one-fifth and one-third of one-fifth of thing from thing. There remain eleven-fifteenths of thing, equal to three dirhems. It is now required to complete the thing. For this purpose, add to it four-elevenths, and do the same with the three dirhems, by adding to them one dirhem and one-eleventh. Then you have four dirhems and one-eleventh, which are equal to thing. This is the sum which is taken out of the debt.

On another Species of Legacy.

"A man dies, leaving his mother, his wife, and two brothers and two sisters by the same father and mother with himself; and he bequeaths to a stranger one-ninth of his capital."

Computation:* You constitute their shares by taking them out of forty-eight parts. You know that if you take one-ninth from any capital, eight-ninths of it will remain. Add now to the eight-ninths one-eighth of the same, and to the forty-eight also one-eighth of them, namely, six, in order to complete your capital. This gives fifty-four. The person to whom one-ninth is bequeathed receives six out of this, being one-ninth of the whole capital. The remaining forty-eight will be distributed among the heirs, proportionably to their legal shares.

If the instance be: "A woman dies, leaving her husband, a son, and three daughters, and bequeathing to a stranger one-eighth and one-seventh of her capital;" then you constitute the shares of the heirs, by taking them out of twenty.* Take a capital, and subtract from it one-eighth and one-seventh of the same. The remainder is, a capital less one-eighth and one-seventh. Complete your capital by adding to that which you have already, fifteen forty-one parts. Multiply the parts of the capital, which are twenty, by forty-one; the product is eight hundred and twenty. Add to it fifteen forty-one parts of the same, which are three hundred: the sum is one thousand one hundred and twenty parts. The person to whom one-eighth and one-seventh were bequeathed, receives one-eighth and one-seventh of this. One seventh of it is one hundred and sixty, and one-eighth one hundred and forty. Subtracting this, there remain eight hundred and twenty parts for the heirs, proportionably to their legal shares.

On another Species of Legacies, viz.

If nothing has been imposed on some of the heirs,* and something has been imposed on others; the legacy amounting to more than one-third.

It must be known, that the law for such a case is, that if more than one-third of the legacy has been imposed on one of the heirs, this enters into his share; but that also those on whom nothing has been imposed must, nevertheless, contribute one-third.

Example: "A woman dies, leaving her husband, a son, and her mother. She bequeaths to a person two-fifths, and to another one-fourth of her capital. She imposes the two legacies together on her son, and on her mother one moiety (of the mother's share of the

*It appears in the sequel (p. 96) that a widow is entitled to $\frac{1}{8}$th, and a mother to $\frac{1}{6}$th of the residue; $\frac{1}{8}+\frac{1}{6}=\frac{7}{24}$; leaving $\frac{17}{24}$ of the residue to be distributed between two brothers and two sisters; that is, $\frac{17}{48}$ between a brother and a sister; but in what proportion these 17 parts are to be divided between the brother and sister does not appear in the course of this treatise. Let the whole capital of the testator $=1$ and let each 48th share of the residue $=v$; $1-\frac{1}{9}=\frac{8}{9}$; $\therefore\ \frac{8}{9}=48v$; $\therefore\ v=\frac{1}{54}$; that is, each 48th part of the residue $=\frac{1}{54}$th of the whole capital.

*A husband is entitled to $\frac{1}{4}$th of the residue, and the sons and daughters divide the remaining $\frac{3}{4}$ths of the residue in such proportion, that a son receives twice as much as a daughter. In the present instance, as there are three daughters and one son, each daughter receives $\frac{1}{5}$ of $\frac{3}{4}$, $=\frac{3}{20}$, of the residue, and the son, $\frac{6}{20}$. Since the stranger takes $\frac{1}{8}+\frac{1}{7}=\frac{15}{56}$ of the capital, the residue $=\frac{41}{56}$ of the capital, and each $\frac{1}{20}$th share of the residue $=\frac{41}{56}\times\frac{1}{20}=\frac{41}{1120}$ of the capital. The stranger, therefore, receives $\frac{15}{56}=\frac{300}{1120}=\frac{15}{56}$ of the capital.

*The problems in this chapter may be considered as belonging rather to Law than to Algebra, as they contain little more than enunciations of the law of inheritance in certain complicated cases.

residue); on her husband she imposes nothing but one-third, (which he must contribute, according to the law)."*

Computation: You constitute the shares of the heritage, by taking them out of twelve parts: the son receives seven of them, the husband three, and the mother two parts. You know that the husband must give up one-third of his share; accordingly he retains twice as much as that which is detracted from his share for the legacy. As he has three parts in hand, one of these falls to the legacy, and the remaining two parts he retains for himself. The two legacies together are imposed upon the son. It is therefore necessary to subtract from his share two-fifths and one-fourth of the same. He thus retains seven twentieths of his entire original share, dividing the whole of it into twenty equal parts. The mother retains as much as she contributes to the legacy; this is one (twelfth part), the entire amount of what she had received being two parts.

Take now a sum, one-fourth of which may be divided into thirds, or of one-sixth of which the moiety may be taken; this being again divisible by twenty. Such a capital is two hundred and forty. The mother receives one-sixth of this, namely, forty; twenty from this fall to the legacy, and she retains twenty for herself. The husband receives one-fourth, namely, sixty; from which twenty belong to the legacy, so that he retains forty. The remaining hundred and forty belong to the son; the legacy from this is two-fifths and one-fourth, or ninety-one; so that there remain forty-nine. The entire sum for the legacies is, therefore, one hundred and thirty-one, which must be divided among the two legatees. The one to whom two-fifths were bequeathed, receives eight-thirteenths of this; the one to whom one-fourth was devised, receives five-thirteenths.

If you wish distinctly to express the shares of the two legatees, you need only to multiply the parts of the heritage by thirteen, and to take them out of a capital of three thousand one hundred and twenty.

On another Species of Legacies.

"A man dies, and leaves four sons and his wife; and bequeaths to a person as much as the share of one of the sons less the amount of the share of the widow."

Divide the heritage into thirty-two parts. The widow receives one-eighth,* namely, four; and each son seven. Consequently the legatee must receive three-sevenths of the share of a son. Add, therefore, to the heritage three-sevenths of the share of a son, that is to say, three parts, which is the amount of the legacy. This gives thirty-five, from which the legatee receives three and the remaining thirty-two are distributed among the heirs proportionably to their legal shares.

If he leaves two sons and a daughter,* and bequeaths to some one as much as would be the share of a third son, if he had one; then you must consider, what would be the share of each son, in case he had three. Assume this to be seven, and for the entire heritage take a number, one-fifth of which may be divided into sevenths, and one-seventh of which may be divided into fifths. Such a number is thirty-five. Add to it two-sevenths of the same, namely, ten. This gives forty-five. Herefrom the legatee receives ten, each son fourteen,

*If the bequests stated in the present example were charged on the heirs collectively, the husband would be entitled to $\frac{1}{4}$, the mother to $\frac{1}{6}$ of the residue: $\frac{1}{4}+\frac{1}{6}=\frac{5}{12}$; the remainder $\frac{7}{12}$ would be the son's share of the residue; but since the bequests, $\frac{2}{5}+\frac{1}{4}=\frac{13}{20}$ of the capital, are charged upon the son and mother, the law throws a portion of the charge on the husband.

*A widow is entitled to $\frac{1}{8}$th of the residue; therefore $\frac{7}{8}$ths of the residue are to be distributed among the sons of the testator. Let x be the stranger's legacy. The widow's share $=\frac{1}{8}(1-x)$; each son's share $=\frac{1}{4}\cdot\frac{7}{8}(1-x)$; and a son's share, minus the widow's share $=\frac{7}{32}-\frac{4}{32}(1-x)=x$; $\therefore x=\frac{3}{35}$.

*A son is entitled to receive twice as much as a daughter. Were there three sons and one daughter, each son would receive $\frac{2}{7}$ths of the residue. Let x be the stranger's legacy. Then $\frac{2}{7}(1-x)=x$; $\therefore x=\frac{2}{9}$; $1-x=\frac{7}{9}$. Each Son's share $=\frac{2}{7}\cdot\frac{7}{9}=\frac{2}{9}$. The Daughter's share $=\frac{1}{7}\cdot\frac{7}{9}=\frac{1}{9}$. The Stranger's legacy $=\frac{2}{9}$.

and the daughter seven.

If he leaves a mother, three sons, and a daughter, and bequeaths to some one as much as the share of one of his sons less the amount of the share of a second daughter, in case he had one; then you distribute the heritage into such a number of parts as may be divided among the actual heirs, and also among the same, if a second daughter were added to them.* Such a number is three hundred and thirty-six. The share of the second daughter, if there were one, would be thirty-five, and that of a son eighty; their difference is forty-five, and this is the legacy. Add to it three hundred and thirty-six, the sum is three hundred and eighty-one, which is the number of parts of the entire heritage.

On Completement.

"A woman dies and leaves eight daughters, a mother, and her husband, and bequeaths to some person as much as must be added to the share of a daughter to make it equal to one-fifth of the capital and to another person as much as must be added to the share of the mother to make it equal to one-fourth of the capital."*

Computation: Determine the parts of the residue, which in the present instance are thirteen. Take the capital, and subtract from it one-fifth of the same, less one part, as the share of a daughter: this being the first legacy. Then subtract also one-fourth, less two parts, as the share of the mother: this being the second legacy. There remain eleven-twentieths of the capital, which, when increased by three parts, are equal to thirteen parts. Remove now from thirteen parts the three parts on account of the three parts (on the other side), and you retain eleven-twentieths of the capital, equal to ten parts. Complete the capital, by adding to the ten parts as much as nine-elevenths of the same; then you find the capital equal to eighteen parts and two-elevenths. Assume now each part to be eleven; then the whole capital is two hundred, each part is eleven; the first legacy will be twenty-nine, and the second twenty-eight.

Computation of Returns.

On Marriage in Illness.

"A man, in his last illness, marries a wife, paying (a marriage settlement of) one hundred dirhems, besides which he has no property, her dowry being ten dirhems. Then the wife dies, bequeathing one-third of her property. After this the husband dies."*

Computation: You take from the one hundred that which belongs entirely to her, on account of the dowry, namely, ten dirhems; there remain ninety dirhems, out of which she has bequeathed a legacy. Call the sum given to her (by her husband, exclusive of her dowry) thing; subtracting it, there remain ninety dirhems less thing. Ten dirhems and thing are already in her hands; she has disposed of one third of her property, which is three dirhems and one-third, and one-third of thing; there remain six dirhems and two-thirds plus two-thirds of thing, the moiety of which, namely, three dirhems and one-

*Let x be the stranger's legacy; a widow's share of the residue is $\frac{1}{8}$th; $\frac{7}{8}(1-x)$ is the residue. $\frac{1}{8}[1-x]$, to be distributed among the children. Since there are 3 sons, and 1 daughter, $\frac{1}{7}\cdot\frac{7}{8}(1-x)$ is a son's share. Were there 3 sons and 2 daughters, a daughter's share would be $\frac{1}{8}\cdot\frac{8}{8}(1-x)$. The difference $= \frac{1}{7}\cdot\frac{7}{8}(1-x) - \frac{1}{8}(1-x) = x$; $\therefore x = \frac{45}{336}$.

*In this case, the mother has $\frac{2}{13}$; and each daughter has $\frac{1}{13}$ of the residue. $1 - x - y = 13v$; $1 - [\frac{1}{5} - v] - [\frac{1}{4} - 2v] = 13v$; i.e. $1 - \frac{1}{5} + v - \frac{1}{4} + 2v = 13v$.

*Let s be the sum, including the dowry, paid by the man, as a marriage settlement; d the dowry; x the gift to the wife, which she is empowered to bequeath if she pleases. She may bequeath, if she pleases, $d+x$; she actually does bequeath $\frac{1}{3}[d+x]$; the residue is $\frac{2}{3}[d+x]$, of which one half, viz. $\frac{1}{3}[d+x]$ goes to her heirs, and the other half reverts to the husband. $\therefore$ the husband's heirs have $s-[d+x]+\frac{1}{3}[d+x]$; and since what the wife has disposed of, exclusive of the dowry, is x, twice which sum the husband is to receive, $s - \frac{2}{3}[d+x] = 2x$; $\therefore \frac{3}{8}[3s-2d] = x$. But $s = 100$; $\therefore x = 35$; $d + x = 45$; $\frac{1}{3}[d+x] = 15$. Therefore the legacy which she bequeaths is 15, her husband receives 15, and her other heirs, 15. The husband's heirs receive $2x = 70$.

third plus one-third of thing, returns as his portion to the husband.* Thus the heirs of the husband obtain (as his share) ninety-three dirhems and one-third, less two thirds of thing; and this is twice as much as the sum given to the woman, which was thing, since the woman had power to bequeath one-third of all which the husband left;* and twice as much as the gift to her is two things. Remove now the ninety-three and one-third, from two-thirds of thing, and add these to the two things. Then you have ninety-three dirhems and one-third equal to two things and two-thirds. One thing is three-eighths of it, namely, as much as three-eighths of the ninety-three and one-third, that is, thirty-five dirhems.

On Emancipation in Illness.

"Suppose that a man on his death-bed were to emancipate two slaves; the master himself leaving a son and a daughter. Then one of the two slaves dies, leaving a daughter and property to a greater amount than his price."* You take two-thirds of his price, and what the other slave has to return (in order to complete his ransom). If the slave die before the master, then the son and the daughter of the latter partake of the heritage, in such proportion, that the son receives as much as the two daughters together. But if the slave die after the master, then you take two-thirds of his value and what is returned by the other slave, and distribute it between the son and the daughter (of the master), in such a manner, that the son receives twice as much as the daughter; and what then remains (from the heritage of the slave) is for the son alone, exclusive of the daughter; for the moiety of the heritage of the slave descends to the daughter of the slave, and the other moiety, according to the law of succession, to the son of the master, and there is nothing for the daughter (of the master).

It is the same, if a man on his death-bed emancipates a slave, besides whom he has no capital, and then the slave dies before his master.

If a man in his illness emancipates a slave, besides whom he possesses nothing, then that slave must ransom himself by two-thirds of his price. If the master has anticipated these two-thirds of his price and has spent them, then, upon the death of the master, the slave must pay two-thirds of what he retains.* But if the master has anticipated from him his whole price and spent it, then there is no claim against the slave, since he has already paid his entire price.

"Suppose that a man on his death-bed emancipates a slave, whose price is three hundred dirhems, not having any property besides; then the slave dies, leaving three hundred dirhems and a daughter." The computation is this:* Call the legacy to the slave

*In other cases, as appears from pages 92 and 93, a husband inherits one-fourth of the residue of his wife's estate, after deducting the legacies which she may have bequeathed. But in this instance he inherits half the residue. If she die in debt, the debt is first to be deducted from her property, at least to the extent of her dowry (see the next problem.)

*When the husband makes a bequest to a stranger, the third is reduced to one-sixth. Vide p. 137.

*From the property of the slave, who dies, is to be deducted and paid to the master's heirs, first, two-thirds of the original cost of that slave, and secondly what is wanting to complete the ransom of the other slave. Call the amount of these two sums p; and the property which the slave leaves u. Next, as to the residue of the slaves' property: First. If the slave dies before the master, the master's son takes $\frac{2}{3}[u-p]$; the master's daughter $\frac{1}{3}[u-p]$; and the slave's daughter $\frac{1}{2}[u-p]$. Second. If the slave dies after the master; the master's son is to receive $\frac{2}{3}p$, and the master's daughter $\frac{1}{3}p$; and then the master's son takes $\frac{1}{2}[u-p]$, and the slave's daughter $\frac{1}{2}[u-p]$.

*The slave retains one-third of his price; and this he must redeem at two-thirds of its value; namely at $\frac{2}{3}\times\frac{1}{3}=\frac{2}{9}$ of his original price.

*Let the slave's original cost be a; the property which he dies possessed of u; what the master bequeaths to the slave, x. Then the net property which the slave dies possessed of is $u+x-a$. $\frac{1}{3}[u+x-a]$ belongs, by law, to the master; and $\frac{1}{2}[u+x-a]$ to the slave's daughter. The master's heirs, therefore, receive the ransom, $a-x$, and the inheritance, $\frac{1}{3}[u+x-a]$; that is, $\frac{1}{3}[(u+a-x)]$; and on the same principle as the slave, when thing. He has to return the remainder of his price, after the deduction of the legacy, or three hundred less thing. This ransom, of three hundred less thing, belongs to the master. Now the slave dies, and leaves thing and a daughter. She must receive the moiety of this, namely, one half of thing; and the master receives as much. Therefore the heirs of the master receive three hundred less half a thing, and this is twice as much as the legacy, which is thing, namely, two things. Reduce this by removing half a thing from the three hundred, and adding it to the two things. Then you have three hundred, equal to two things and a half. One thing is, therefore, as much as two-fifths of three hundred, namely, one hundred and twenty.

This is the legacy (to the slave,) and the ransom is one hundred and eighty.

"Some person on his sick-bed has emancipated a slave, whose price is three hundred dirhems; the slave then dies, leaving four hundred dirhems and ten dirhems of debt, and two daughters, and bequeathing to a person one-third of his capital; the master has twenty dirhems debts." The computation of this case is the following:* Call the legacy to the slave thing; his ransom is the remainder of his price, namely, three hundred less thing. But the slave, when dying, left four hundred dirhems; and out of this sum, his ransom, namely, three hundred less thing, is paid to the master, so that one hundred dirhems and thing remain in the hands of the slave's heirs. Herefrom are (first) subtracted the debts, namely, ten dirhems; there remain then ninety dirhems and thing. Of this he has bequeathed one-third, that is, thirty dirhems and one-third of thing; so that there remain for the heirs sixty dirhems and two-thirds of thing. Of this the two daughters receive two-thirds, namely, forty dirhems and four-ninths of thing, and the master receives twenty dirhems and two-ninths of thing, so that the heirs of the master obtain three hundred and twenty dirhems less seven-ninths of thing. Of this the debts of the master must be deducted, namely, twenty dirhems; there remain then three hundred dirhems less seven-ninths of thing; and this sum is twice as much as the legacy of the slave, which was thing; or, it is equal to two things. Reduce this, by removing the seven-ninths of thing, and adding them to two things; there remain three hundred, equal to two things and seven-ninths. One thing is as much as nine twenty-fifths of three hundred, which is one hundred and eight; and so much is the legacy to the slave.

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emancipated, is allowed to ransom himself at two-thirds of his cost, the law of the case is that 2 are to be taken, where 1 is given. $\therefore\ \frac{1}{3}[(u+a-x)] = 2x$; $\therefore\ x = \frac{1}{7}[u+a]$; If, as in the example, $u = a$, $x = \frac{2}{7}a$; the daughter's share $= \frac{1}{7}[3u-2a]$; The master's heirs receive $\frac{2}{7}[u+a]$.

*Let the slave's original cost $= a$; the property he dies possessed of $= u$; the debt he owes $= e$. He leaves two daughters, and bequeaths to a stranger one-third of his capital. The master owes debts to the amount h; where $a = 300$, $u = 400$, $e = 10$, $h = 20$. Let what the master gives to the slave, in emancipating him $= x$. Slave's ransom $= a - x$; slave's property $-$ slave's ransom $= u + x - a$; Slave's property $-$ ransom $-$ debt $= u+x-a-e$; Legacy to stranger $= \frac{1}{3}[u+x-a-e]$; Residue $= \frac{2}{3}[u+x-a-e]$; The master, and each daughter, are, by law, severally entitled to $\frac{1}{3} \times \frac{2}{3}[u+x-a-e]$; The master's heirs receive altogether $a-x+\frac{1}{3}[u+x-a-e]$ or $\frac{1}{3}[a-x]+\frac{2}{3}[u-e]$, which, on the principle that 2 are to be taken for 1 given, ought to be made equal to $2x$. But the author directs that the equation for determining x be $\therefore\ x = \frac{3}{25}[7a+2[u-e]-3h] = 108$.

End of the Translation of the Algebra of Mohammed ben Musa.

Notes on the Arabic Text [ملاحظات على النص العربي]

The following brief notes are intended to point out for further enquiry a few circumstances connected with the history of Algebra:

1. The initial words of the treatise, as cited by Haji Khalfa, are: قال ابو عبدالله محمد بن موسى الخوارز ("Abu Abdallah Mohammed ben Musa of Khowarezm said…").
2. The term الجبر (*al-jabr*) literally signifies "restoration" or "completion," referring to the operation of transferring negative quantities to the other side of an equation as positive. The term المقابلة (*al-muqabala*) means "reduction" or "opposition," referring to the process of cancelling like quantities on both sides of an equation.
3. The Arabic word شيء (*shay'*, literally "thing") is used throughout the text as the designation of the unknown quantity—what we should now call x. It corresponds exactly to the *res* ("thing") of the early Italian algebraists, and to the *coss* of the Germans.
4. The word مال (*māl*, literally "property," "wealth," "possession") is used to denote the square of the unknown. It is rendered by Rosen as "square."
5. The word جذر (*jidhr*, literally "root") is used in the same sense as our "root." The word واجب (*wājib*, literally "obligatory") is used for the "absolute number"—what we call the constant term.
6. The method of geometrical demonstration here employed by al-Khwarizmi is worthy of especial notice. It shows that, like the Greek mathematicians, he was not satisfied with merely giving rules, but endeavoured to establish their accuracy by demonstration. His proofs are founded upon the second proposition of Book II of Euclid's Elements, and upon the well-known property of the gnomon.
7. The formula for the circumference of a circle given in the text (Text p. 51, Transl. p. 72) is $c = \frac{62832}{20000}d$, which gives $\pi = 3.1416$. This value was known to the Hindu mathematician Aryabhata, and the circumstance furnishes a strong argument for the opinion that al-Khwarizmi derived part of his information from an Indian source.
8. The comparisons drawn between the Algebra of the Arabs and that of the early Italian writers might perhaps have been more numerous and more detailed; but the enquiry was here restricted by the want of some important works. Montucla, Cossali, Hutton, and the Basil edition of Cardanus' *Ars magna*, were the only sources which the translator had the opportunity of consulting.
9. The solutions which the author has given of the remaining problems of this treatise, are, mathematically considered, for the most part incorrect. It is not that the problems, when once reduced into equations, are incorrectly worked out; but that in reducing them to equations, arbitrary assumptions are made, which are foreign or contradictory to the data first enounced, for the purpose, it should seem, of forcing the solutions to accord with the established rules of inheritance, as expounded by Arabian lawyers.
10. The object of the lawyers in their interpretations, and of the author in his solutions, seems to have been, to favour heirs and next of kin; by limiting the power of a testator, during illness, to bequeath property, or to emancipate slaves; and by requiring

payment of heavy ransom for slaves whom a testator might, during illness, have directed to be emancipated.

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Appendix: Arabic–English Glossary of Algebraic Terms

THE FOLLOWING GLOSSARY presents the complete Arabic mathematical terminology employed by Mohammed ben Musa al-Khwārizmī in his *Kitāb al-Mukhtaṣar fī Ḥisāb al-Jabr wal-Muqābala*, together with Rosen's English equivalents and modern mathematical interpretations.

Arabic Term	Transliteration	Meaning and Usage
	I. CORE ALGEBRAIC TERMS	
الجبر	*al-jabr*	**Restoration, completion.** The operation of transferring negative quantities to the other side of an equation as positive. One of the two fundamental operations of algebra, giving the science its name.
المقابلة	*al-muqābala*	**Reduction, opposition, balancing.** The process of cancelling like quantities on both sides of an equation; also the reduction of equations to standard form.
مال	*māl*	**Square, property, wealth.** The square of the unknown quantity (x^2). Literally "property" or "possession," reflecting the mercantile context of early algebra. Rosen consistently renders this as "square."
جذر	*jidhr*	**Root.** The unknown quantity itself (x); also used for square roots of known numbers. Literally "root," "base," or "origin."
شيء	*shay'*	**Thing.** The unknown quantity (x). The most common designation for the unknown in Arabic algebra. Corresponds to the Latin *res* and the Italian *cosa*.

Continued on next page

Arabic Term	Transliteration	Meaning and Usage
عدد	*‘adad*	**Number.** The absolute or constant term in an equation; a known quantity unrelated to the unknown. Rosen renders this as “simple number” or “number.”
واجب	*wājib*	**Obligatory, requisite.** The constant term or absolute number in certain contexts. Used by later writers; al-Khwārizmī himself uses *‘adad*.
	II. Geometrical Terms	
مربع	*murabba‘*	**Quadrate, square figure.** A geometric square used in demonstrations. Distinguished from *māl* which refers to the algebraic square.
خط	*khaṭṭ*	**Line.** A straight or curved line.
مستقيم	*mustaqīm*	**Straight, rectilinear.** Describes a straight line or straight angle.
قائمة	*qā’ima*	**Perpendicular, cathetus.** A line standing at right angles to another. Literally “standing upright.”
طول	*ṭūl*	**Length.** The longer dimension of a figure.
عرض	*‘arḍ*	**Breadth, width.** The shorter dimension of a figure.
دائرة	*dā’ira*	**Circle.** Literally “that which revolves.”
قطر	*qaṭr*	**Diagonal, diameter.** The line passing through the centre of a figure.
	III. Mercantile and Practical Terms	
درهم	*dirham*	**Dirhem.** A silver coin; the standard monetary unit in algebraic calculations. Derived from the Greek drachma.
ثمن	*thaman*	**Price, value.** The cost of goods in mercantile problems.

Continued on next page

Arabic Term	Transliteration	Meaning and Usage
كيل	*kayl*	**Measure.** A unit of dry measure for grain, etc.
مقدار	*miqdār*	**Quantity, amount.** A general term for an unknown amount.
جملة	*jumla*	**Sum, total.** The aggregate of quantities.
نسبة	*nisba*	**Ratio, proportion.** The relation between two quantities.
	IV. Legal and Legacy Terms	
وصية	*wasiyya*	**Legacy, bequest.** A gift made by will, especially on death-bed. Subject to the restriction that it may not exceed one-third of the estate without heirs' consent.
ميراث	*mīrāth*	**Inheritance, heritage.** The estate left by a deceased person.
فرائض	*farā'iḍ*	**Obligatory shares.** The legally fixed portions of an estate due to prescribed heirs according to Islamic law.
عتق	*'itq*	**Emancipation, manumission.** The freeing of a slave, especially by a master's death-bed declaration.
مكاتبة	*mukātaba*	**Manumission contract.** An agreement whereby a slave purchases freedom by instalments.
مهر	*mahr*	**Dowry, marriage settlement.** The payment made by a groom to the bride, part of which may revert to the husband's heirs.
	V. Book Title and Formulaic Expressions	
كتاب المختصر في حساب الجبر والمقابلة	*Kitāb al-Mukhtaṣar fī Ḥisāb al-Jabr wal-Muqābala*	**The Compendious Book on Calculation by Completion and Reduction.** The full Arabic title of al-Khwārizmī's algebra.

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Arabic Term	Transliteration	Meaning and Usage
بسم الله الرحمن الرحيم	*Bi-smi Llāhi r-Raḥmāni r-Raḥīm*	**In the Name of God, the Gracious, the Merciful.** The traditional opening formula (basmala) of Arabic books.
الحمد لله	*Al-ḥamdu li-Llāh*	**Praise be to God.** The opening formula of the author's preface.

Note on Transliteration. The system of transliteration employed above follows the standard academic convention for Arabic: macrons (ā, ī, ū) denote long vowels; dots beneath consonants (ṭ, ḥ, ṣ, ḍ) denote the Arabic emphatic consonants; the hamza (') represents the glottal stop; and the 'ayn (') represents the voiced pharyngeal fricative.

The Six Canonical Forms of Equations

Rosen's English	**Arabic**	**Modern Form**
Squares equal Roots	المال يساوي الجذور	$ax^2 = bx$
Squares equal Numbers	المال يساوي الأعداد	$ax^2 = c$
Roots equal Numbers	الجذور تساوي الأعداد	$bx = c$
Squares and Roots equal Numbers	المال والجذور يساويان الأعداد	$ax^2 + bx = c$
Squares and Numbers equal Roots	المال والأعداد يساويان الجذور	$ax^2 + c = bx$
Roots and Numbers equal Squares	الجذور والأعداد تساوي المال	$bx + c = ax^2$

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Enhanced Bilingual Edition with Original Arabic Terminology Restored, 2025